War and Structural Transformation Evidence from Germany after World War I

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April 10, 2024

Abstract

This paper investigates the effect of war on structural change by looking at the economic transformation of Germany after WWI. For this, we combine newly geocoded data on the birthplaces of 7.2 million German WWI casualties with county level data from the 1910 and 1925 German business and agricultural census. Based on continuous difference-in-differences estimations, we show that WWI casualties accelerated the moving-out of the primary sector, yet delayed the formation of the service industry. While the agricultural sector responded to the war-induced human capital shock with the absorption of very small farms by medium-sized farms and mechanization, we argue that the educational requirements of the tertiary sector made deceased WWI soldiers harder to replace, delaying the formation of the service and trade industries.

Keywords: WWI, war casualties, structural transformation, economic development.
JEL Classification: O14, N44

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Acknowledgments: We thank Philipp Ager, Antonio Ciccone, Axel Dreher, Andreas Fuchs, Martin Gassebner, Erik Hornung, Jean Lacroix, Markus Ludwig, Felix Rösel, and seminar audiences at the 2019 Beyond Basic Questions Seminar, the 2022 Centre for European Studies Dissertation Workshop at Harvard University, the Spatial Dimensions of the Labour Market at the Institut für Arbeitsmarkt- und Berufsforschung Nuremberg 2022, the Political Economy of Democracy and Dictatorship 2023, the Ruhr Graduate School Doctoral Workshop 2023, the German Development Economics Conference 2023, and the 2023 Beyond Basic Questions Seminar for helpful comments and suggestions. We are grateful to Dr. Jesper Zedlitz of the Verein für Computergenealogie for providing data on the digitized loss lists, to Christoph Halbmeier for sharing data on geocoded tax revenues, to Uli Schubert for sharing the hierarchical list of German administrative regions in 1910, and to De Juan et al. (2023) for sharing the information on dead and wounded WWI soldiers. We thank Erik Schulte, Fynn von Reeken, and Laura Wedemeyer for excellent research assistance. Tobias Korn and Lukas Wellner acknowledge generous support by the German Research Foundation (DFG) as part of the Research Training Group 1723 "Globalization and Development."

1 Introduction

Population shocks can have large and sustained consequences for local economic development. The literature provides ample evidence that sudden labor supply shocks from out-migration (Hornbeck and Naidu, 2014; Andersson et al., 2022), pandemics (Brainerd and Siegler, 2003; Voigtländer and Voth, 2013a; Karlsson et al., 2014), natural disasters (Vigdor, 2008; Boustan et al., 2012) or wars (Voigtländer and Voth, 2013b) can persistently change economic trajectories. Yet, identifying the causal effect of such labor supply shocks is challenging. For instance, natural disasters and wars usually affect the local infrastructure and capital stocks along with the local population. Similarly, we must assume that individuals' decisions to migrate are endogenous to local economic conditions. The closest setting to an exogenous negative population shock that does not directly affect other economic inputs are pandemics. The Black Death, for instance, killed an estimated 40% of the population in the Middle Ages and had vast economic consequences for affected locations (Voigtländer and Voth, 2013a). But can we directly infer from shocks in pre-industrial economies to more modern settings? Whereas pandemics still leave their toll on today's economies, their scale is not comparable to the Black Death. For instance, about 2% of the population died during the Spanish Flu in the early 20th century (Karlsson et al., 2014), and mortality from Covid-19 was significantly below 0.5% of the population in most countries (Johns Hopkins University, 2022). Very likely, these later pandemics do not constitute a human capital shock as transitory as the Black Death to understand how industrialized economies respond to vast human capital reductions.

We analyze another, modern human capital shock that had severe demographic consequences: German casualties during World War I.[1] During the four years of war, an estimated two million German soldiers (about 15% of the male population between 20 and 45) died, and many more were wounded or incapacitated. Notable in our setting is the fact that, different from most other violent settings, the shock did barely affect the capital stock of the economy through war-related destruction.[2] Except for a short period at the beginning of the war, all combat took place outside of German territory.

At the end of the 19th century, Germany was in the midst of the second wave of industrialization. In 1882, the primary sector was still by far the largest employee with 48.4% of the workforce employed in agriculture, while the secondary sector share was 29.6%, and the tertiary sector made up for 22.0% of the workforce. In the wake of WWI, primary employment was at 27.4%, secondary employment at 40.9%, and tertiary employment at 31.7% .

A host of labor-saving and labor-assisting technologies became available that transformed the routines of manual labor (Atack et al., 2019). Women started entering the regular workforce in large numbers, often taking up new-tasks opened up by the de-skilling of labor brought by the new technologies (Brown and Philips,

[1] We use the term casualty as those who died in war.

[2] See Broadberry and Harrison (2005) for an estimate of the war-related destruction of physical assets in the combatant countries. The authors report a 3.1% rate of destruction of German domestic assets as a share of prewar assets.

1986). In addition, the large-scale promotion of higher education transformed Germany's workforce and society. A new form of White Collar workers emerged. These were well educated and high-skilled individuals who took over lower managerial tasks within firms, ending the aspirations of less educated Blue Collar workers to climb up the ranks with more on-the-job experience (Kocka, 1981). These White Collar Workers were referred to as "Angestellte" vis-á-vis the Blue Collar "Arbeiter." They received better working contracts with higher pay, better job security, and more vacation days and turned into Germany's new social upper middle class.

This paper investigates how the war-induced labour force shock affected the structural transformation in Germany at the local level. For this, we use newly geocoded data on the birthplaces of 7.2 million German soldiers from the official casualty lists published by the German army during the war. We combine this data with newly digitized German business and agriculture census waves from 1907 and 1925, which provide detailed sector level data at the county level. Among other things, we observe the number of employees across 18 sector categories as well as agricultural landholdings by size before and after the war.

To investigate counties' trends in structural transformation, we leverage continuous difference-in-differences estimations following Callaway et al. (2021) to identify the average causal response of a marginal increase in a location's casualty rate on various outcomes. Our identification approach relies on the assumption of "conditional strong local parallel trends." We therefore need to assume that, conditional on covariates, locations with marginally different treatment "doses" would have followed the same trend absent the treatment. For this assumption to hold, we introduce treatment quartile fixed effects in all our regressions. This is, we always estimate the marginal effect of an increase in the treatment dose among counties with a similar treatment intensity. Therefore, our identification approach allows for selection into higher or lower treatment intensities, i.e., quartiles, but demands a quasi-random distribution of treatment doses within each quartile.

In addition, we further account for local institutional and economic factors that may have influenced the casualty rate indirectly via differences in drafting ratios or the deployment to more or less lethal battle scenes of local regiments. Tests for parallel pre-trends suggest that the "conditional strong local parallel trends assumption" is likely to hold for our treatment variable.

Our results show that counties with more war casualties exhibited lower employment growth, especially in the primary and tertiary sector. In the industrial sector, war casualties did not affect the employment trend. This is because firms in this sector recruited unskilled workers from the informal daily laborer as well as small-scale farming sectors to rapidly fill vacant positions. The agricultural sector also reduced employment, despite cultivating constant amounts of land. This was possible by medium-sized farms investing in mechanization and replacing small subsistence farms. The tertiary sector, finally, faced a significant decline in employment as high-skilled White Collar Workers could not be replaced by machines or labor reserves. Hence, counties with more war fatalities saw a decline in the tertiary sector and therefore the new upper middle class of White Collar Workers.

Our findings contribute to three strands of the literature. First, we contribute

to the literature on the effects of negative labor supply shocks. In terms of technology adoption, Hornbeck and Naidu (2014) and Andersson et al. (2022) find that areas more affected by out-migration have higher rates of technology adoption in agriculture. For individual-level outcomes, Goldin (1991) and Goldin and Olivetti (2013) find that the labor supply of women increased due to male mobilization in World War II. Acemoglu et al. (2004) show that this increase in female labor force participation lowered wages because female labor became a better substitute for male labor after the war. In a closely related setting to our study, Boehnke and Gay (2020) and Brodeur and Kattan (2022) show that French casualties in World War I as well as US casualties in World War II, respectively, led to higher female labor force participation after the war. Gay (2023) confirms the positive effect of WWI casualties on female labor force participation in France, and additionally shows that the excess labor demand improved attitudes towards working women. Closer to our paper, Eder (2022) finds that farmers moved into low-skilled manufacturing jobs in Austria where casualties from World War II were higher. Finally, Huber et al. (2021) investigate the expulsion of Jewish Managers in 1930s Germany and find similar to us that highly educated workers were difficult to replace, hence forcing an overall deterioration of production. Our findings extend this line of the literature by providing a more detailed look into within-sector changes, shedding light on economic transformations within the agricultural sector, and providing first evidence for the decline of tertiary employment related to human capital shocks.

Second, we contribute to a growing literature which examines the impact of WWI in the interwar period. Acemoglu et al. (2022) use Italian WWI casualties to argue that the death toll in the war was a major cause for the rise of communism in Italy, which in turn caused the counter-movement and rise of the right-wing government led by Mussolini in the mid 1920s. The impact of local exposure to the war on the rise of the Nazi party in Germany is examined in a similar fashion by Koenig (2023) and De Juan et al. (2023). Both papers establish a direct connection between WWI casualties or the return of veterans to their homes and increased vote shares of extreme right and the Nazi Party in Germany. They attribute these effects to the organizational skill of right-wing veterans' organizations and the common experience of suffering and human loss associated with casualties.

Third, several recent studies rely on the raw data set of German WWI casualties from Verein für Computergenealogie (2019a) that we also use in our analysis. For instance, De Juan et al. (2023) investigate the effect of casualties in a county on voting behavior in the post-war era. Ciccone (2021) uses casualty list entries as a negative, exogenous local population shock for the state of Württemberg. Kersting and Wolf (2021) use the names and birthplaces of soldiers to identify patterns in first names as signs of nation-building after German unification in 1871. All these studies rely on the original geocoding provided by Computergenealogie (2019a), which we improve by providing a new and more accurate geocoding of the data source on the county level for more than 84% of observations in the data set, as we discuss in Appendix A.

This paper proceeds as follows. section 2 provides a brief historical background and theoretical considerations. section 4 presents the data used. section 5 describes

the identification strategy. section 6 presents the results and related robustness tests. section 7 discusses the mechanisms. Finally, section 8 concludes.

2 Historical Background

WWI was the first fully modern global war spanning vast regions across Europe, the Middle East, Africa, the Pacific, and Asia.[3] WWI ranks among the most deadly wars in history, with an estimate of 10 million dead soldiers. From July 1914 to November 1918, the conflict persisted, pitting the Central Powers—comprising Germany, Austria-Hungary, and the Ottoman Empire—against the Allies, which included Great Britain, France, Russia, Italy (joining in 1915), and the United States (joining in 1917), among manz other nations.

After the envisioned German military strategy of a quick victory against France failed, the Western front consisted of trench warfare from the English channel in the north to Switzerland in the South. The Eastern front was more mobile, yet despite heavy casualties, there was no victory on either side, until the Russian Empire collapsed and Soviet Russia dropped out of the war. The war continued in the West, and Allied forced secured victory in the summer of 1918. As a result of the war, the Ottoman Empire and Austria-Hungary were broken up. Germany was demilitarized, ceded 12% of its pre-war territory, and faced large reparation payments to the Allies.[4]

The German war effort involved almost universal mobilization of men aged 17-45. In total, 13 million soldiers fought for the German Empire during the war. Casualty numbers were extremely high, making WWI one of the most deadliest wars in human history. The scale of the war, but also its brutality using heavy artillery, machine guns, and chemical warfare to inflict unprecedented numbers of casualties on the enemy, let contemporaries in shock. The German imperial army alone lists two million dead soldiers.[5] From those who returned from the war, a total of six million soldiers suffered minor or major injuries, not accounting for those who grappled with severe psychological trauma from the war exposure.

WWI had vast economic consequences. After the war, the national income had fallen by a third. Production numbers shrunk significantly, especially in mining (17%), iron and steel (37%), construction (65%), textiles (83%), and cereals (43%) (Ritschl, 2005). Labour productivity fell, both in the civilian and military sector, and real income per capita shrunk by 15–25% (Ritschl, 2005). German structural transformation slowed down significantly. The primary sector transition stagnated, and so did the formation of the industries and service sectors. As a consequence,

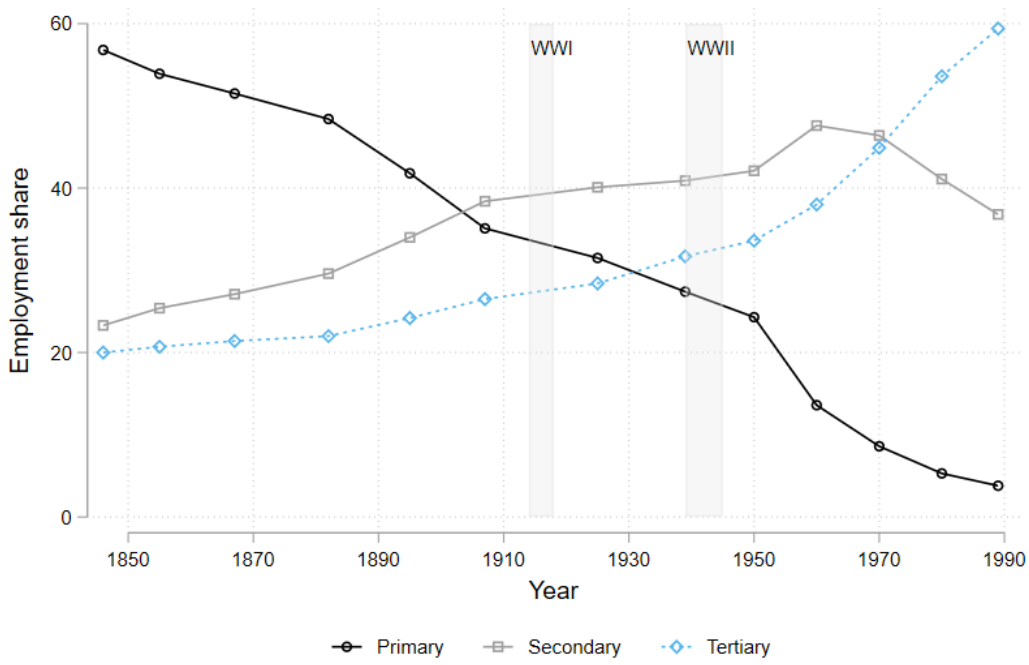
[3]In this paper, we focus on the European war theater and in particular on the events related to the German war effort.

[4]Reparations largely consisted of direct monetary payments as well as deliveries of coal, steel, and other industrial goods. While monetary payments were made out of central government funds, we believe it unlikely that in kind reparation payments correlated with the WWI casualty rates. Similarly, it is unlikely that the 1923 hyperinflation correlated systematically with the casualty rate.

[5]Rates where similar across the main European belligerents, with even higher casualty rates in France and Russia.

the German economy still employed 24% of its workforce in agriculture, compared to only 5% in Great Britain (Broadberry, 1997).[6] Structural transformation would only return to its pre-war levels after 1950 (see Figure 1). The next chapter provides theoretical considerations on how the WWI casualties contributed to the slowed down structural transformation of the German economy.

Figure 1: WWI and the structural transformation in Germany



Note: This graph illustrates the sectoral composition of employment in Germany across the census waves in 1855, 1867, 1882, 1895, 1907, 1925, 1939, and 1950. Data from Rahlf (2022).

3 Theoretical Considerations

This paper investigates how the German economy — in particularly German businesses — reacted to the war-induced human capital shock experienced during WWI. For this, we focus on the two defining aspects of the classical structural transformation: the transition of the workforce from the agricultural sector to industries and services, and the associated increase in demand for a more specialized, high-skilled workforce. We theorize that the unprecedented death toll — more than two million soldiers, about 15% of the male population between 20 and 45, had died in the trenches of WWI — slowed down the German structural transformation in Germany, in particularly in sectors that relied more strongly on high-skilled labour.

[6]The structural change of the workforce out of the relatively large and unproductive primary sector is understood a large contributor to Germany's post WWII economic boom (Kindleberger, 1967; Temin, 2002; Braun and Kvasnicka, 2014).

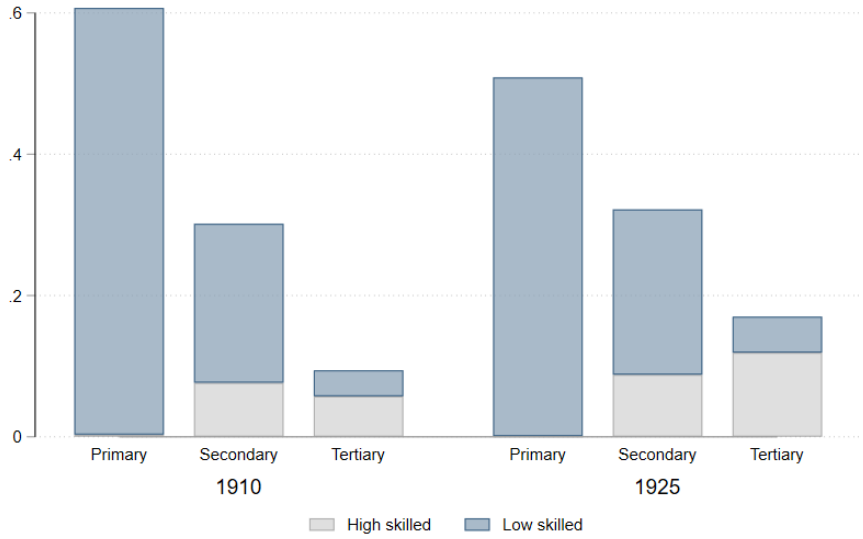
German army mobilization was universal. In expectancy of a short war, mobilization rates — and subsequently casualty rates — were high across workers from all sectors of the economy, even in the war-relevant industries. Rasch (2022) estimates that even in the industrial centers of the Ruhr, an average of 30% of the workforce were recruited at the beginning of the war. Mobilization did not discriminate based on the skill set of the recruits. Factory workers as well as their manager were drafted and served during the war. This changed to some extent when the prospect of a short war vanished and the German economy turned towards war production. If anything, then, mobilization rates among low-skilled, industrial workers were lower as these workers were speared to maintain war production.

German businesses faced significant challenges due to the high number of casualties during World War I, which led to a notable reduction in the overall workforce. The process of replacing deceased workers varied depending on their skill level. For low-skilled laborers, who could be trained on the job, finding replacements was relatively straightforward, often sourced from other industries. However, replacing jobs that required high skill levels posed a much greater challenge, primarily because their roles involved intricate tasks that demanded years of education and formal training. As war recruitment did not spear high-skilled workers, high-skilled labor became scarce, and firms had to compete for the remaining employees. This made the consequences of WWI more severely felt in sectors that relied on high-skilled employment.

The tertiary sector had the highest human capital requirements. With an increase in the specialization of firms also came an increase in the demands of skills of the labour force from unskilled, manual workers to high skilled employees conducting administrative, professional, or scientific tasks (Clark, 1940; Kuznets, 1973; Kocka, 1981; Feinstein, 1999). This was particularly the case in the German economy, where educational focus was high since the early 19th century (Kocka, 1981). The share of high skilled workers in services in 1910 was high (60.1% in 1910, as shown in Figure 2), and the dependence on high skilled work made the tertiary sector particularly vulnerable to the loss of human capital. What is more, the tertiary sector was in direct competition with the war-benefiting industries for the remaining pool of high-skilled employees, yet could not offer the same wages. War losses therefore resulted in vacancies. This is likely to have hampered the development of the tertiary sector in counties with higher WWI casualties and these counties fell behind in the structural transformation process of the German economy after WWI.

The secondary sector strongly relied on low-skilled labour for repetitive, semi-automated tasks. While the administration of large businesses required to some extend also high-skilled labour predominantly for administrations and legal tasks, low-skilled worker comprised of 74.6% of the total workforce in the secondary sector in 1910. What is more, parts of the industrial sector relevant for war production benefited from the war. Those industries received vast public investments, faced high public demand, and had priority access to raw materials to continue production. Wages for skilled and unskilled workers in these sectors continued to rise with employment on high levels (Ritschl, 2005). After the first months of the war, casualty rates were lower in war-related industries, and if workers did as soldiers in war,

Figure 2: Employment and skill level by sector, 1910 and 1925



Notes: This figure depicts the share of high-skilled workers and low-skilled workers of total employment by sector according to the 1910 and 1925 business census.

the high wages facilitated new hires — crowding out low-skill employment in the non-war industries, in agriculture, and services, and high-skilled employment in the tertiary sector. We therefore do not expect a large effect of WWI casualties on the secondary sector, in particular among industries that were relevant to fight the war.

The agricultural sector comprised of low-skilled workers undertaking physically demanding jobs in a sector was still at the beginning of the mechanisation process. While farmers themselves were exempt, a large share of WWI soldiers were agricultural workers. This made the agricultural sector the sector which was most strongly exposed to WWI (Afflerbach, 2018). What is more, the agricultural sector strong competition for the remaining pool of low skilled workers, who were offered higher wages in the industrial sector. This further exacerbated the effects of the war-related human capital shock in agriculture. Farmers had several options to respond to the increased capital/labor ratio: mechanization, extensification, or reduction of the amount of land cultivated. Here, we expect farms to mechanize where mechanization was not fully undertaken (in particular in family owned, medium-sized farms), to extensify where mechanization was already implemented (the case of larger estates), or not possible due to insufficient financial means or lands (small farms).

The next section presents the data we use to test these hypothesises.

4 Data

To measure the war-induced human capital shock on structural transformation in Germany, we collect a vast set of data on German WWI casualties and German

economic statistics from before and after the war. Our unit of observation is the 1907 county level.[7]

Casualty data. We derive our main explanatory variable from the number of WWI casualties based on the official war records of the German Imperial Army, the so called loss lists ("Verlustlisten"). Digitized by Verein für Computergenealogie (2019a), these lists contain 8.5 million entries with information on each soldier's name, birthplace, rank, regiment and the type of casualty reported from wounded to dead.[8] For this study, we geocoded the birthplaces of 7.2 million WWI soldiers at the county level.[9]

Systematic information on the casualty status – that is, whether the entry constitutes a dead or wounded soldier – has been digitized by De Juan et al. (2023). In combination with the population census of 1907, we calculate the share of total casualties in the 1907 male population. Figure 3 shows the distribution of the casualty rate, along with the army districts (black lines) and regiment districts (white lines) within which military drafting and deployment was organized. As depicted by the map, casualty rates tended to be higher in more rural areas, such as East Prussia and West Prussia, Pomerania, Posen, and Silesia, and lower in more urban areas such as Berlin, Danzig, the Ruhr area as well as Southern Bavaria.

Census data. We newly digitized German census data from the German business census waves from 1907 and 1925.[10] The census reports information on company level employment for all companies in Germany. It provides this information for the primary, secondary, and tertiary sector, as well as subcategories ("Gewerbegruppe"). We matched the data at the level of sub-sectors across the two census waves, and merged them with our stable 1907 county boundaries.[11] The census data contain information on the number of businesses as well as employees and worker per county

[7]The county ("Kreis") constitutes Germany's fourth administrative layer. We match all data to constant 1907 county borders. To do so, we proceed in two steps. First, we link all counties whose borders remain unchanged from the first year of observation in 1907 until our last year of observation in 1933. Second, for counties that split or merge over time but where the outside boundaries remain unchanged, we trace all administrative reforms and aggregate these counties to the biggest common spatial entity. This provides us with 725 stable county units. For the remaining 229 counties, we were unable to link them over time due to exhaustive re-drawings of their boundaries amid large scale administrative reforms, especially in Baden and Thuringia.

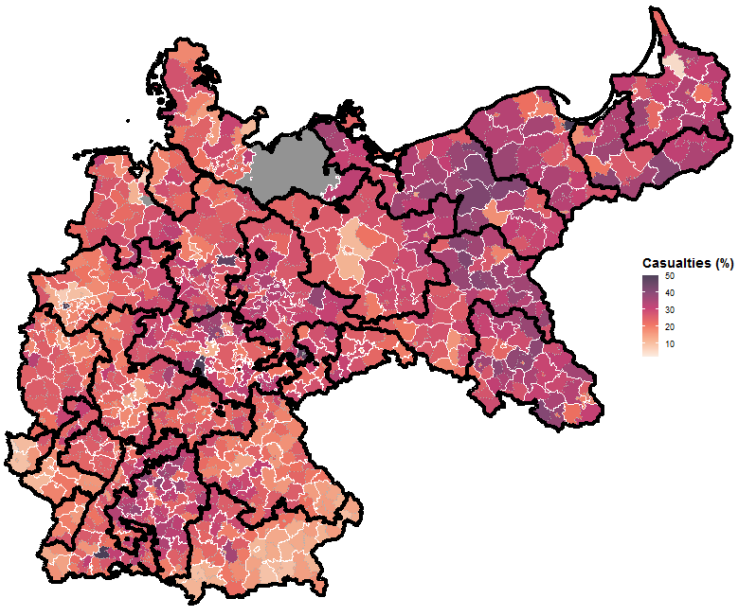
[8]This corresponds to 65% of the soldiers that fought during WWI. Note that not all of this information is reported for all entries, and only information on soldiers' names and birthplaces have been digitized systematically.

[9]To this end, we exploited the hierarchical structure of loss list entries that typically consist of the name of the municipality of birth, followed by the county or administrative district and matching birthplace entries with a list of German municipality names in 1910 from Schubert (2020). The remaining uncoded 1.3 million entries were of too broad resolution to be coded on the county level (e.g. "Bayern"), inconclusive (e.g., "Neustadt," of which there exist more than 40 in Germany), or outside the 1910 German boundaries (e.g., "Zürich"). We describe the geocoding process as well as improvements made vis-à-vis the original coding provided by Verein für Computergenealogie (2019a) in detail in Appendix A.

[10]We use the business census as compared to the employment census as the former has more detailed information on the number of businesses, sector level employment, blue and white collar workers, and mechanization.

[11]We provide detailed information on the different sector classifications and how they are matched in the Table B2.

Figure 3: WWI casualty rate



Note: This map depicts the number of WWI casualties as share of the 1910 male population at the county level. Thick black lines depict army district borders. White borders depict regiment district borders. Thin dotted gray lines depict county borders. Grey areas denote missing data.

and sector. Both census waves also indicate the level of skill of employees by distinguishing high-skilled, white-collar workers ("Angestellte") from low-skilled, blue collar workers ("Arbeiter"). From this, we code the share of white collar workers in total employment.

We digitized data from the German agricultural census in 1907 and 1925. The census contains data on the total number of farms by county, the total amount of plotted land in hectare, and the number of farm workers employed in a county. As is common in German agricultural censuses, it provides this information separately for five different farm sizes: very small (0 – 2 ha), small (2 – 5 ha), medium sized (5 – 20 ha), large (20 – 100 ha), and very large (above 100 ha). From the 1925 census, we further leverage data on agricultural machine usage for four major types of agricultural machinery: threshing machines, mowers, seeding machines, and cultivators.

5 Identification Strategy

To measure the war induced labour force shock on the local economic conditions in Germany, we rely on continuous difference-in-differences estimations that exploit the institutional setup of the German military recruitment and follow Callaway et al. (2021) by comparing counties within similar casualty rates.

Contemporaneous accounts from WWI suggest that a soldiers survival in bat-

tle was largely determined by chance. Large scale industrial warfare, in particular machine guns and heavy artillery, lead to an unprecedented high level of casualties, with up to 50% of the German WWI casualties caused by heavy artillery fire (Reichswehrministerium, 1934). As Remarque (1928) reports: "Above all is chance. When a bullet is coming, all I can do is duck. I can neither know nor steer where it lands."

While a soldiers fate during battle was largely random, the number or total casualties across battles during the war largely differed. For the German army, casualty numbers peaked during the 1914 invasion of France, the battles at Verdun and at the Somme in 1916, and the Spring Offensive in 1918.[12] A soldiers fate during war then therefore depended on the deployment to specific battlefields.

German army deployment during WWI was organized by army district. Each of these districts was responsible for staffing and the equipment of an army. The German Empire was divided into 25 army districts. Each of these army districts was further divided into several regiment districts, which were responsible for recruitment and staffing of specific regiments.[13] Commonly, a regiment district comprises of three or four counties.[14] The idea of this setup was to fill regiments with soldiers from neighbouring districts to increase morale and fighting spirit. Part of a specific army, these regiments fought together in the same war theaters and in consequence faced comparable likelihoods to die in war.

Army deployment is conditional on mobilization. And while mobilization was organized within the very same regiment districts, certain individuals were exempt from military service. This was for instance the case of workers in the war-related industries. Mobilization thus differed depending on the economic structure of the county. Unfortunately, county level mobilization rates do no longer exist. We therefore use a proxy, which models the number of absent men during war times. To track the labour toll of the war, the German government undertook a wartime census in 1916, which we use to obtain data on the gender ratio in each county. We argue that counties with a higher female share as compared to the prewar period are likely to have a higher mobilization rate.[15]

We argue that two men from counties with a similar mobilization rate and of the same regiment district had a comparable chance to survive the war. We provide evidence for this in Table 1. For this, we simply regress the share of casualties soldiers on various pre-war county characteristics in a cross-sectional OLS regression. We include the economic structure of the county controlling for primary, secondary, and tertiary employment, as well as the share of employees working in war industries. We also include an indicator for urban areas as political and economic conditions in denser populated areas likely affected mobilization from these places. Finally, we

[12]Figure C4 provides an overview of the development of the German casualty numbers from 1914 to 1919.

[13]Figure B1 a) shows the borders of the army districts, Figure B1 b) shows the borders of the regiment districts.

[14]Figure B2 shows the distribution of the number of counties per military district.

[15]The census provides information for the male population in three categories: civilian males, male military personnel, and foreign prisoners of war. We exclude prisoners of war because they were not part of the local male population and we exclude soldiers because they were either on leave or in local training camps at the time of the census.

separately control for whether or not the county belongs to the state of Prussia, as Prussia was the central power in the military administration, and Prussian officers might have deemed Prussian regiments to be more reliable due to higher loyalty to the regime and the traditionally higher military spirit in Prussian counties.

As can be seen in column 1, these variables are meaningful determinants of the number of WWI casualties. Casualty numbers were higher in counties with a higher employment in the primary and secondary sector, and lower in counties with higher employment in tertiary sector. Furthermore, counties with a higher share of employees in the war industries experienced lower casualty shares. In line with our expectation, casualty numbers were also higher in counties belonging to Prussia. Also note that the included control variables include 27.0 percent of the variation in the casualty variable. When including the controls for deployment and mobilization, this value increases to 72.7 percent and the control variables lose explanatory power. Yet, some county level characteristics remain important predictors of the casualty rates despite controlling for mobilization and deployment. Importantly, agricultural areas report higher casualty rates, while urban and more industrialized counties report lower rates. To avoid that our estimates are biased, we thus introduce a third central control variable in column 3.

To address remaining differences across counties within the same regiment district, and similar mobilization rates, we follow Callaway et al. (2021) and differentiate into different treatment intensities. While differences between counties with high and low casualty rates might be driven by differences in the economic, social, and political conditions in these counties, we argue that within a close range of similar doses of the treatment, the selection into a specific dose is arguably random. For this, we categorize the sample of counties into four groups of treatment intensities. Our identifying variation thus no longer comes from comparing counties with high and low casualty rate, but counties within the range of similar casualty rates. Within units of the same dose, local parallel trends are then sufficient to interpret our results as causal local marginal effects Callaway et al. (2021).

Column 3 of Table 1 shows this approach at work. Beyond deployment and mobilization, we now also include the casualty quartiles. These casualty quartiles divide all counties in four groups of similar treatment doses, from low to high casualty rates. Figure C3 illustrates the distribution of the treatment intensity for the defined quartiles.[16] We now only compare counties within neighbourhoods of similar treatment doses. As can be seen in column 3, the introduction of these quartiles renders all previously significant predictors of the casualty rate insignificant. Now, neither the total number of workers in the agricultural sector, nor the share of workers engaged in war industries has predictive power on the number of WWI casualties. Also note that within these doses, our setting now explains 89.4 percent of the variation in the casualty rate. We argue that controlling for treatment intensity via casualty quartiles, army deployment, and army mobilization allows us to interpret our findings as causal.

Based on this, we estimate the following continuous difference-in-differences es-

[16]Figure B3 in the Appendix shows the geographic distribution of casualty quartiles.

Table 1: Determinants of the casualty rate

	(1)	(2)	(3)
Employment: primary	0.009	0.006	0.001
	(0.001)***	(0.002)***	(0.001)
Employment: secondary	0.006	-0.002	-0.001
	(0.002)***	(0.002)	(0.001)
Employment: tertiary	-0.006	-0.002	0.000
	(0.003)**	(0.004)	(0.002)
Employment share: war industries	-0.051	-0.056	-0.015
	(0.013)***	(0.020)***	(0.010)
Population	-0.004	-0.003	0.003
	(0.004)	(0.006)	(0.003)
Urban	-0.003	-0.004	-0.001
	(0.003)	(0.003)	(0.002)
Prussia	0.018	-0.014	-0.025
	(0.003)***	(0.022)	(0.019)
Observations	775	722	722
R^2	0.290	0.687	0.894
Mean DV	0.099	0.099	0.099
Regiment district	.	✓	✓
Mobilization	.	✓	✓
Casualty quartiles	.	.	✓

Notes: Results from cross-section OLS regressions. The outcome variable is the share of casualties in the male 1910 population. "Employment: primary" indicates the logged number of agricultural workers 1910. "Employment: secondary" indicates the logged number of industrial workers in 1910. "Employment: tertiary" indicates the logged number of service workers in 1910. "Employment share: war industries " indicates the share of workers working in war industries in 1910. "Population" measures logged 1910 population. "Urban" indicates a county population above 75,000 or a population density above 150. "Prussia" indicates that a county belongs to the state of Prussia. Standard errors clustered by county in parentheses, Conley standard errors in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

timation:

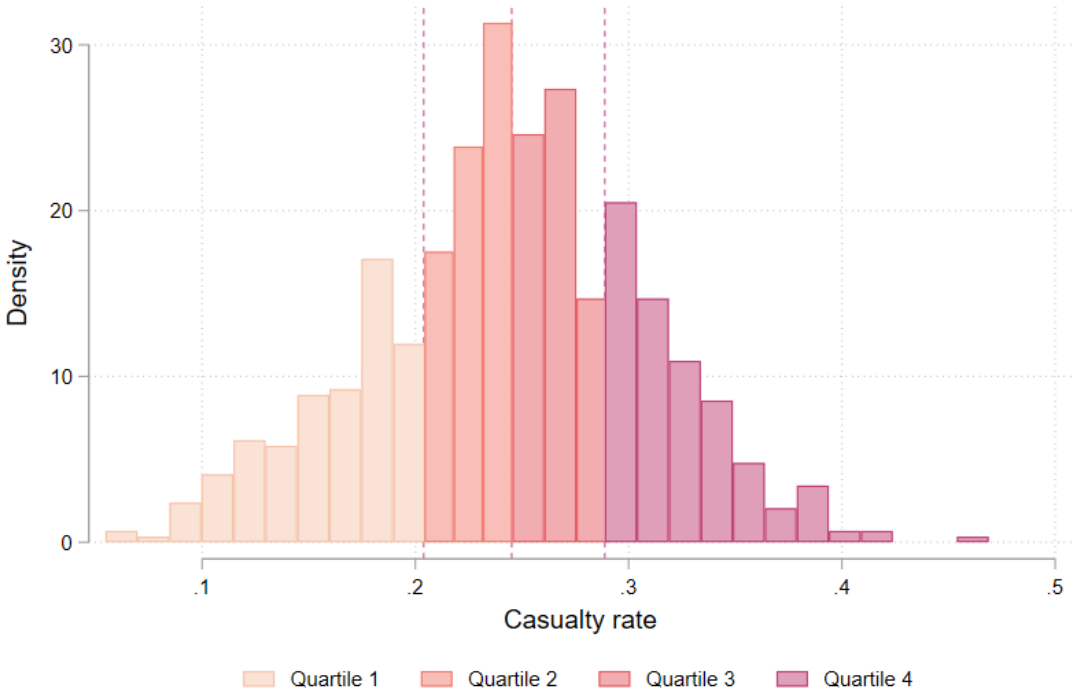
$$Y_{i,t} = \beta_k Casualty_i^k \times P_t + P_t + \gamma_i \times P_t + \delta_i \times P_t + \zeta D_i^k \times P_t + \eta M_i \times P_t + \theta Pop_{i,t} + \epsilon_{i,t} \quad (1)$$

where we observe our outcome variable $Y_{i,t}$ for each county i across two time periods t, with $t = 0$ referring to the pre-war period (usually 1907), and $t = 1$ referring to the post-war period (usually 1925).

We observe each county i's treatment intensity $Casualty_i$, which we measure as the share of a county's casualties over its 1907 male population. Our causal parameter of interest is β_k, the coefficient of the interaction between each county i's treatment intensity $Casualty_i^k$ and an indicator variable for observations after the war (P_t).

Our estimations follow the common two-way fixed-effects setup, with the period-indicator P_t accounting for counties' common trends over time, and the county-fixed effects γ_i controlling for time-invariant heterogeneity across counties. $\delta_i \times P_t$ indicate regiment district fixed effects. In addition to these fixed effects, we control for

Figure 4: WWI casualty rates and casualty quartiles



Notes: This figure illustrates the empirical distribution of our main explanatory variable, the rate of WWI casualties in the pre-war 1910 male population, along with the allocation into quartiles for our identification of local marginal effects in a histogram. We show the geographic distribution of these quartiles across counties in Figure B3.

casualty quartile fixed effects D_i^k, interacted with the post-war indicator P_t. These quartile fixed effects help establish the strong local parallel trends assumption by reducing the overall variation within the dose to variation within a neighborhood of similar doses D_i^k.[17] M_i indicates the excess female population share as proxy for mobilization, and $Pop_{i,t}$ indicates county population over time. $\epsilon_{i,t}$ denotes the error term.

Based on the regiment district fixed effects, mobilization rates, and casualty rate quartile fixed effects provide us with a setting where we compare counties with different casualty rates among counties with a similar exposure to WWI.

6 Results

Table 2 shows the results following equation 1. We first look at the effect of WWI casualties on overall county level employment, and then on employment in the primary, secondary, and tertiary sector.

[17]For this, we calculate treatment specific quartiles based on the treatment shares $k \in \{total, dead\}$.

Table 2: WWI casualties and structural transformation

	Employment (1)	Employment: primary (2)	Employment: secondary (3)	Employment: tertiary (4)
Casualties × post	-4.65428	-2.03624	-1.39253	-4.18979
	(1.27049)***	(0.74355)***	(2.05606)	(1.74993)**
	[1.51514]***	[0.58249]***	[1.19627]	[0.94258]***
County FE	✓	✓	✓	✓
Post x regiment district FE	✓	✓	✓	✓
Post x mobilization	✓	✓	✓	✓
Post x quartile FE	✓	✓	✓	✓
Observations	1432	1432	1432	1432

Notes: Results from continuous difference-in-differences estimations following equation 1. Outcome variables measure logged total employment (column 1), and logged employment in the primary, secondary, and tertiary sector (columns 2–4). "Casualties × post" indicates the share of casualties in the 1907 male population interacted with a dummy indicating the post-war period. All specifications control for logged population. Standard errors clustered by county in parentheses, Conley standard errors in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We observe a reduction in overall employment in counties with a higher casualty rate during WWI. This reduction in employment was driven by the primary and tertiary sector, while we do not observe an effect of WWI casualties on secondary sector employment. The effects are sizeable. A one percentage point increase in the casualty rate reduces total employment by 4.65%. This reduction is most prominent in the tertiary sector, where employment decreases by about 4.19% for a one percentage point higher death rate. We also find a significant reduction in agricultural employment, where a 1% increase in the fatality rate leads to 2.04% fewer workers.

For the interpretation of the results, we have to remember the overall change in the German economy from 1907 to 1925 (see Figure 1). First, primary employment was in decline, yet at significantly slower rates than in the pre-war period. In counties with higher casualty rates, this decline was faster, than in counties with fewer casualties. Thus, the war-induced labour force shock increased economic transition in agriculture, and slowed down the transition in counties with fewer casualties. Second, secondary employment was reaching its pre-war heights, yet the growth of the secondary sector slowed down after the war. Our results suggest that this was not directly related to the casualties of the war. Third, the formation of the tertiary sector continued after WWI, but again at reduced pace. We observe that WWI casualties played a significant role in the slower emergence of the tertiary sector. We explore the mechanisms behind this in section 7.

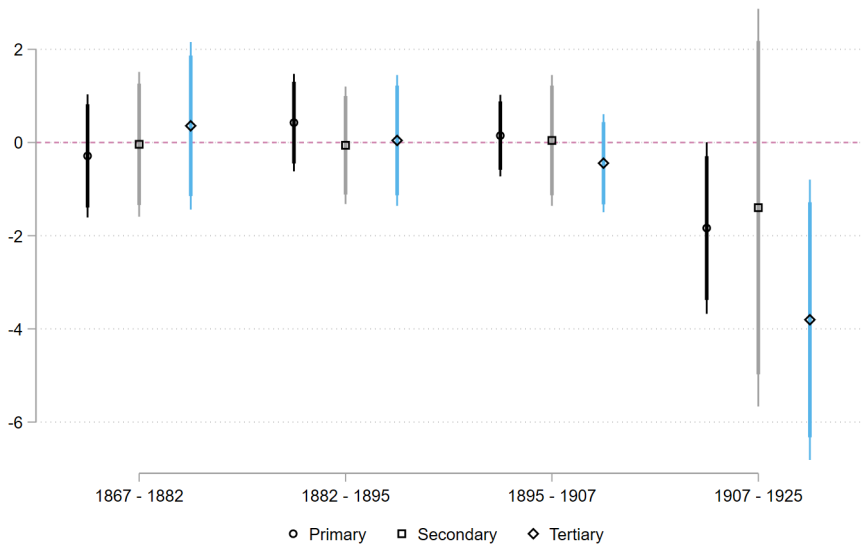
We test the robustness of these findings in several ways. First, we test for local parallel trends among counties with similar treatment doses. We investigate pre-trends in the pre-sample period using sector-level employment data for Prussian counties in 1867, 1882, 1895, and 1907 from Galloway (2007).[18] If counties with

[18]We have to restrict this analysis to Prussia as reliable data for the remaining German states for the same time periods is not available. To account for administrative reforms that occurred during this 60 year panel, we harmonize the counties to a constant sample over time, aggregating county mergers and splits to the biggest common entity.

similar casualty shares followed otherwise parallel growth trajectories during the war, we should not see any significant differences across counties in the periods before the eventual war casualties were realized.

Figure C6 reports the results from continuous difference-in-differences regressions that follow our main specification in equation 1, but for the time periods 1867–1882, 1882–1895, and 1895–1907.[19] We find no evidence of different employment trajectories across counties with marginally different casualty rates. Also, our baseline results hold when only examining the subset of Prussian counties as we show in Figure C6. The results are correspond to our baseline estimates in coefficient size.

Figure 5: WWI casualties and structural transformation, pre-trends



Notes: This figure depicts the coefficients as well as their 90%- and 95% confidence intervals from the test for pre-trends by estimating the continuous difference-in-differences regressions as outlined in equation 1, but for the periods 1867–1882, 1882–1895, and 1895–1907 for the harmonized sample of Prussian counties in the Galloway data. "Primary" indicates logged employment in the agriculture sector, colored in black. "Secondary" indicates logged employment in the manufacturing sector, colored in gray. "Tertiary" indicates logged employment in the trade and service sector, colored in blue. We report results by pairs of years for the three sectors as indicated on the X-axis. All coefficients derive from separate regressions. All specifications include army district fixed effects, the female share, casualty quartiles, and logged population. Data before 1910 derive from Galloway (2007).

Our results also hold when excluding specific subsamples, as we show in Table C4. Column 1 reports our main results for comparison. Column 2 drops counties in the Ruhr area that were occupied by French and Belgium forces from 1923–1925. After Germany's failure to meet the reparation requirement, France and Belgium occupied the territory and confiscated industrial goods and raw material, which is likely to have stunted the industrial development in the area and therefore affect our results.

[19]Table C1 in the Appendix presents the corresponding results in table form.

Column 3 drops the East Prussian territory invaded by Russian forces in 1914, which was the only 1914 Germany territory exposed to fighting. Column 4 restricts the sample to the non-Prussian counties to show that results also hold when excluding the central state within the German Empire.[20] Columns 5 and 6 drop the highest decile of counties with the highest respectively lowest casualty rate to control that our results are not driven by possible outliers. Convincingly, our results hold when dropping these subsamples.

7 Mechanisms

Counties with higher casualty rates in WWI experienced an accelerated transformation of the primary sector and a delayed tertialization. First, we turn to the primary sector to understand the mechanisms behind these results.

Throughout the 19th century, agricultural employment has been on the decline, driven mostly by labour saving technologies such as the threshing machine. Such innovations freed up labour force, and allowed farmers to cultivate more land with fewer labourers. This trend continued in the early 20th century, and the primary share of total employment in 1925 was down to 31.5%, as we show in Figure B6. This decrease was strongest among farms of 20 hectares and more, while there was an increase in employment of very small, subsistence farms. The number of farms changed only marginally, increasing from 4.16 million farms in 1910 to 4.23 million farms in 1925. As with the number of workers, the number of very small farms expanded, while in particular the number of large and very large estates decreased. Also the total cultivated land in Germany shrunk from 23,037 thousand hectares in 1910 to 21,738 thousand hectares in 1925.[21] Most of the reduction in farm land occurred at large estates, while very small to medium sized farms expanded.

WWI heavily drew on the labour supply of the agricultural sector. While farmers themselves where exempt from conscription, their workers where not, and conscription reduced the availability of farm workers. Farmers could increase their own work effort where possible, substitute workers with labour-saving technology, or reduce farm sizes in response to this war-induced labour force shock. We test farmers' response in Table 3, using logged primary sector employment (panel A), logged farm size (panel B), and the logged number of farms (panel C) as an outcome variable in equation 1. In panel D, we additionally look at mechanization based on a cross-sectional OLS regression using 1925 threshing machine usage as the outcome variable. For all of these outcomes, we observe the outcome by farm size from very small farms below two hectares (column 2) to very large farms with more than 100 hectares (column 6). This differentiation allows us to understand how the agricultural sector coped with the WWI-induced labour force shock.

After WWI, small-scale farming increased in Germany. Yet, in regions with

[20] The strong reduced sample size prevents us from including regiment district fixed effects in this regressions and we use army district fixed effects instead.

[21] The reduced area corresponds to 13 thousand hectares, an area equivalent to the extend of the modern city states of Hamburg and Berlin. Note that all these numbers are based on our final sample and thus do not include land ceded in 1919.

higher casualty rates, this occurred significantly less. In these counties, fewer people worked in small-scale farming, and the number of small-scale farms and the total cultivated land on them decreased. The increase in small-scale farming occurred in counties with low casualty rates, and we hypothesise that indeed war returnees are behind this increase in smallholder farming, which had a high capability of absorbing low-skilled workers after the war. For farms between 2-5 ha, we observe no change. Rather, medium-sized, often family-owned farms consolidated their land, arguably at the expense of the very small farms. Interestingly, they did so without expanding their labour force. Results from panel D suggest that they compensated the war-related labour shortage with labour saving technology, such as threshing machines. Large farms, traditionally relying on a larger workforce to cultivate their lands, found the number of workers further reduced, and, as their amount of land remained constant, also relied more strongly on labour saving technologies in 1925.

Overall, agricultural transformation accelerated in counties with higher casualty rates after WWI. We observe land consolidation in the hands of mid-sized farms and higher labour-saving technology usage after the war, while counties with lower WWI casualties experienced an increase in agricultural employment, in particular in small scale farming. We now turn to the secondary and tertiary sector.

Secondary employment was reaching its pre-WWII plateau with about 40.1% of total employment in 1925. Our main results in Table 2 show that there was no effect of WWI casualties on secondary employment. Table 4 further shows that there was no effect on either war-related, or non-war related industries (panel A). Further, there was no significant change in the share of high-skilled employees in the secondary sector. The required skills to work in the secondary sector, in particular of factory workers in standardized production, was relatively low. As shown in Figure 2, three quarters of the employees in the secondary sector in 1910 were low-skilled workers. Their tasks would be typically physically demanding or repetitive with few specific skills required.

The results confirm our hypothesis that, even when high-skilled workers died in war, companies in the secondary sector were able to replace these workers, arguably from further qualifying the large pool of unskilled labourers (panel B). Finally, we observe a reduction in the number of firms in the secondary sector. The decease, as expected, occurred entirely in the non-war-related industries, who strongly affected by the adverse economic conditions during wartime. Further versions of this paper will analyze this further. Finally, and based on 1925 cross-sectional data, we do not observe a change in technology adaption in the secondary sector, based on the share of more modern electric engines (as compared to less modern wind, water, or heat power generators) on the total power generated by the firms.

Table 3: WWI casualties, employment, and landholdings in the primary sector

	Total	Very Small <2ha	Small <5ha	Medium <20ha	Large <100ha	Very large >100ha
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Log employment						
Casualties × post	-2.48567	-4.20384	-0.35615	0.21245	-4.03054	-0.34757
	(0.84270)***	(1.49288)***	(0.77889)	(0.66622)	(1.69134)**	(4.02021)
	[0.72129]***	[1.46061]***	[0.37448]	[0.50650]	[1.15144]***	[2.57811]
Mean DV	9.287	7.918	7.511	8.021	6.746	4.538
Panel B: Log farm size						
Casualties × post	-0.41424	-2.49938	0.37125	1.07104	-1.64098	-4.17911
	(0.38629)	(0.87139)***	(0.53331)	(0.61516)*	(1.79530)	(5.31818)
	[0.24731]*	[0.83132]***	[0.24034]*	[0.50156]**	[1.26130]	[3.86883]
Mean DV	9.796	6.968	7.647	8.780	8.025	5.776
Panel C: Log number of farms						
Casualties × post	-2.25306	-4.08647	0.22588	0.79961	-1.21961	-0.92552
	(0.99237)**	(1.59770)**	(0.51896)	(0.61207)	(0.74957)	(1.28062)
	[0.95333]***	[1.59957]***	[0.26138]	[0.43152]*	[0.45143]***	[0.69060]
Mean DV	8.287	7.587	6.460	6.543	4.618	1.853
Post x regiment district	✓	✓	✓	✓	✓	✓
Post x mobilization	✓	✓	✓	✓	✓	✓
Post x quartiles	✓	✓	✓	✓	✓	✓
Observations	1,431	1,431	1,431	1,431	1,431	1,431
Panel D: Ownership of threshing machines (cross section)						
Dead	15.01375	-0.45315	9.97514	17.12852	14.56019	9.69485
	(7.29256)**	(5.06850)	(7.15286)	(7.82572)**	(7.27274)**	(6.16819)
	[7.81648]*	[4.52229]	[6.88716]	[8.30168]**	[8.45651]*	[6.20097]
Mean DV	6.527	2.583	4.740	5.885	4.535	2.433
Regiment district	✓	✓	✓	✓	✓	✓
Mobilization	✓	✓	✓	✓	✓	✓
Quartiles	✓	✓	✓	✓	✓	✓
Observations	396	396	396	396	396	396

Notes: Panel A–C show results from continuous difference-in-differences regressions following Equation 1. Outcome variables measure the logged number of employees in the agricultural sector (panel A), the logged farm size in hectares (panel B), and the logged number of farms (panel C). Columns distinguish between farm sizes, ranging from very small farms (column 2) to very large commercial farms (column 6). "Dead × post" indicates the share of fatalities in the 1910 male population interacted with a dummy indicating the post-war period. Panel D shows results from cross-section OLS regressions. "Dead" indicates the share of fatalities in the 1910 male population. All specifications control for logged population. Standard errors clustered by county in parentheses, Conley standard errors in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: WWI casualties, employment, and mechanization in the secondary sector

	(1) Industry	(2) War-related industry	(3) Other industry
Panel A: total employment			
Dead × post	-1.39253	3.62237	-2.30076
	(2.05606)	(3.75847)	(2.19474)
Mean DV	8.358	5.906	8.128
Panel B: high-skilled employment share			
Dead × post	-0.16365	-0.03965	-0.10746
	(0.28542)	(0.03175)	(0.06980)
Mean DV	0.311	0.013	0.060
Panel C: number of firms			
Dead × post	-3.17644**	0.39765	-3.25119**
	(1.51410)	(2.13714)	(1.54711)
Mean DV	6.897	4.319	6.767
Panel D: average firm size			
Dead × post	1.48353	5.39790**	0.68463
	(1.13329)	(2.40761)	(1.10241)
Mean DV	1.690	2.032	1.609
Post x regiment district	✓	✓	✓
Post x mobilization	✓	✓	✓
Post x quartiles	✓	✓	✓
Observations	1,432	1,432	1,432
Panel D: mechanization (cross section)			
Death rate	-0.20486	0.67491	-0.35896
	(0.54451)	(1.02495)	(0.54832)
Mean DV	0.424	0.607	0.398
Panel E: female labour force participation (cross section)			
Death rate	-5.28165	-7.86791	-5.39784
	(3.21048)	(4.60006)*	(3.15316)*
Mean DV	6.860	4.111	6.726
Regiment district	✓	✓	✓
Mobilization	✓	✓	✓
Casualty quartiles	✓	✓	✓
Observations	720	720	720

Notes: Panel A–C show results from continuous difference-in-differences regressions following Equation 1. Outcome variables measure the logged number of employees in the industrial sector (panel A), the logged number of high-skilled employees in the industrial sector (panel B), and the logged number of firms (panel C). Column 1 reports the results for the entire industrial sector, column 2 for war-related industries, and column 3 for all other industries. See Table B2 for the respective categorization of sectors. "Dead × post" indicates the share of fatalities in the 1910 male population interacted with a dummy indicating the post-war period. Panel D shows results from cross-section OLS regressions. "Dead" indicates the share of fatalities in the 1910 male population. All specifications control for logged population. Standard errors clustered by county in parentheses, Conley standard errors in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: WWI casualties, employment, and the tertiary sector

	(1) Services	(2) Services: transport	(3) Services: consumption
Panel A: total employment			
Dead × post	-4.18979**	-2.24700	-3.69730**
	(1.74993)	(3.04407)	(1.64802)
Mean DV	7.419	5.494	7.210
Panel B: high-skilled employment share			
Dead × post	-0.90985**	-1.09252	-0.89232**
	(0.38004)	(0.73864)	(0.34834)
Mean DV	0.720	0.544	0.764
Panel C: number of firms			
Dead × post	-3.64560**	0.59576	-4.02385***
	(1.47736)	(2.00708)	(1.49984)
Mean DV	6.674	4.161	6.569
Panel D: average firm size			
Dead × post	-0.59243	-4.39048*	0.15246
	(0.65244)	(2.55898)	(0.48104)
Mean DV	1.150	1.684	1.076
Post x regiment district	✓	✓	✓
Post x mobilization	✓	✓	✓
Post x quartiles	✓	✓	✓
Observations	1,432	1,432	1,432

Notes: Panel A–C show results from continuous difference-in-differences regressions following Equation 1. Outcome variables measure the logged number of employees in the tertiary sector (panel A), the logged number of high-skilled employees in the tertiary sector (panel B), and the logged number of firms in the tertiary sector (panel C). Column 1 reports the results for the entire tertiary sector, column 2 only for the subcategory trade, and column 3 only for the subcategory transport. "Dead × post" indicates the share of fatalities in the 1910 male population interacted with a dummy indicating the post-war period. Standard errors clustered by county in parentheses, Conley standard errors in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Finally, we turn to the tertiary sector, which was on a steady rise since the 1880s, and made up for 28.4% of total employment in 1925. Our results suggest that tertiary sector growth was systematically lower in counties with a higher WWI death toll. We provide further details to this in Table 5, distinguishing the transport sector (arguably among the services in strong demand), from all other related services, which mostly comprise of consumption goods, such as trades, insurances, gastronomy, barbers, and pharmacies. Columns 2 presents the results for transport services, column 3 for consumption services. As we expected, the decrease in tertiary sector employment derives from a reduction in employment in consumption goods, while we do not observe a significant decrease in transport-related employment in counties with higher casualty rates.

Next, we look at high-skill employment. Work in the trades and services sector required specific skills, often acquired through specialized educational institutions

and formal training (Kocka, 1981). Consequently, more than two thirds of employees in the tertiary sector was high skilled (see Figure 2). In contrast to primary or secondary sector employment, these skills are difficult to transfer across sectors, and employees thus difficult to replace. As we show in column 1, the share of high-skilled employment indeed decreased in counties with higher employment. Again, the effect is stronger is driven in the consumption related services, while there is not decile in the transport sector. The same holds true for the number of firms, which declined in the tertiary sector overall, and in particular in the consumption related services, while there is even a positive, yet insignificant change in the number of companies engaged in transport.

8 Conclusion

This paper examines how the loss of life during World War I affected the local structural transformation in Germany. The German context of WWI provides an exceptional case study to investigate the effect of a severe population shock in isolation from other economic destruction that usually accompanies such shocks. Due to the fact that fighting mostly took place outside of German territory, the infrastructure and capital stock were left largely untouched. Hence, we can identify the sole effect of the population shock on various economic outcomes.

For our analysis, we collected, combined, and harmonized various unique data sets. First of all, we geocoded more than 8.5 million casualty list entries. What is more, we digitized several censuses and archival data on local economic outcomes, and harmonized them across the German territory to account for various reforms to Germany's subnational borders. This allowed us to not only investigate national trends in different economic outcomes, but to also estimate how counties that suffered more under WWI casualties deviated from this national trend in a Continuous Difference-in-Differences setting. Our identification approach relies on a "Conditional Strong Local Parallel Trends" assumption following Callaway et al. (2021), which requires a quasi-random allocation of treatment intensities among units in similar areas of treatment intensity, conditional on covariates. We show that, conditional on treatment quartile fixed effects and institutional determinants of war deployment and conscription, pre-war economic characteristics lose predictive power of local war exposure. Additionally, we observe no evidence of non-parallel economic trajectories before the war.

Our results demonstrate that the labor supply shock induced by WWI casualties had significant effects on counties' structural transformation. Most importantly, we find that harder hit counties fall behind in the transition towards tertiary production and face lower growth rates of White Collar workers, who were the emerging upper middle class at the turn of the century. Investigating mechanisms, we find plausible evidence that these effects accrued due to a lower substitutability of White Collar workers, on which especially the tertiary sector relied. Blue Collar workers in industry could be replaced by agricultural workers and labor reserves from the informal sector, while a loss in agricultural workers could be cushioned by more rapid mechanization. Only in the tertiary sector, where production relied on specific tasks

and highly trained individuals, job vacancies remained unfilled.

Our findings shed new light on the harm political violence inflicts on transforming economies. Structural transformation requires establishing new modes of production, which increasingly rely on specific human capital that cannot easily be re-produced once lost. Our findings hence bear important insights for developing countries that are currently undergoing structural transformation: socio-political shocks like civil wars, disease, or brain drain complicate the move to modern production technologies as highly trained employees are both essential and hard to replace once lost.

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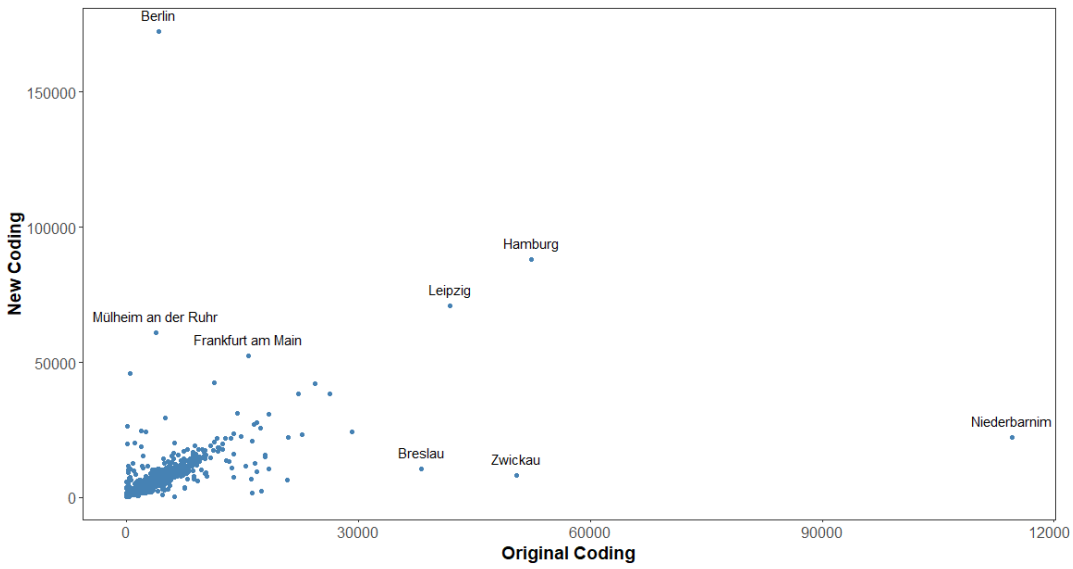
A Geocoding of Casualties

We derive the casualty data as our main explanatory variable from front reports, the so-called "Verlustlisten" or loss-lists, which list the names, ranks, regiments, and birthplaces of soldiers that were wounded or killed during battle. In a crowd-sourcing effort, the Verein für Computergenealogie (2019a) manually digitized the birthplaces of the 8.5 million entries in these lists. Our geocoding process relies on a string-matching algorithm, which compares the manually digitized location names with a list of communities based on Germany's administrative structure of 1910, i.e., four years before the start of the war, collected by Schubert (2020).

The original entries of soldiers' birthplaces follow no specific structure. The majority of entries consists of only one word, e.g., "Berlin," or a combination of two words separated by a comma, e.g., "Spandau, Berlin." In some instances, an entry contains up to five separate words, all separated by a comma. Unfortunately, these comma-separated location descriptions do not follow a specified structure. In many instances, the first entry identifies a town or village, while the later entries identify higher administrative entities. In several instances however, this order may be reversed, e.g., to "Berlin, Spandau." What is more, information on a higher local level does not always identify an official administrative entity, but local information on the region or the next bigger town. Examples for these are entries of the form "Calw, Schwarzwald" or "Heiligenstadt, Mühlhausen." Such cases complicate a direct matching of entries and our list of communities.

Figure A1: Differences in geocoding



Note: This graph illustrates the differences in geocoded casualties between the original coding and our coding. On the horizontal axis, we plot the casualties per county according to the original coding, while the vertical axis displays the casualties per county according to our new coding.

Therefore, we proceeded in the following steps. First, we separated all entries by the comma to attach all distinct location information for each entry. Next, we assigned each list-entry to an entry in our community list, starting from the highest

order of information. Our community list contains, for each community, information at five administrative levels, i.e., the "Land," "Provinz," "Bezirk," "Kreis," and "Gemeinde." For each casualty list entry, we started with the last information of the entry's string, for example with "Baden" in the entry "Neuenheim, Heidelberg, Baden." For this information, we started by looking for matches in the highest administrative level, i.e., the "Land," and then moving down to lower levels until a direct match was found. In the example here, we would directly find a match for "Baden" as a "Land." We would then proceed with the second part of information, looking at lower administrative levels in our community list, though subsetted to the region we already identified.

This direct pattern matching involved a number of difficulties. First, several entries were either misspelled or incorrectly typed during the digitization process. Therefore, we corrected numerous entries where we could not identify a direct match. Second, several entries would not result in an exact match. For example, the entry "Mühlhausen" could either refer to the town Mühlhausen near Erfurt in what is today the state of Thuringia, or to the district Mulhouse, which is part of the state or Alsace which today is in France. We worked through all instances of inexact matches manually, sorting the entries to the respective administrative area based on each entry's lower level information.

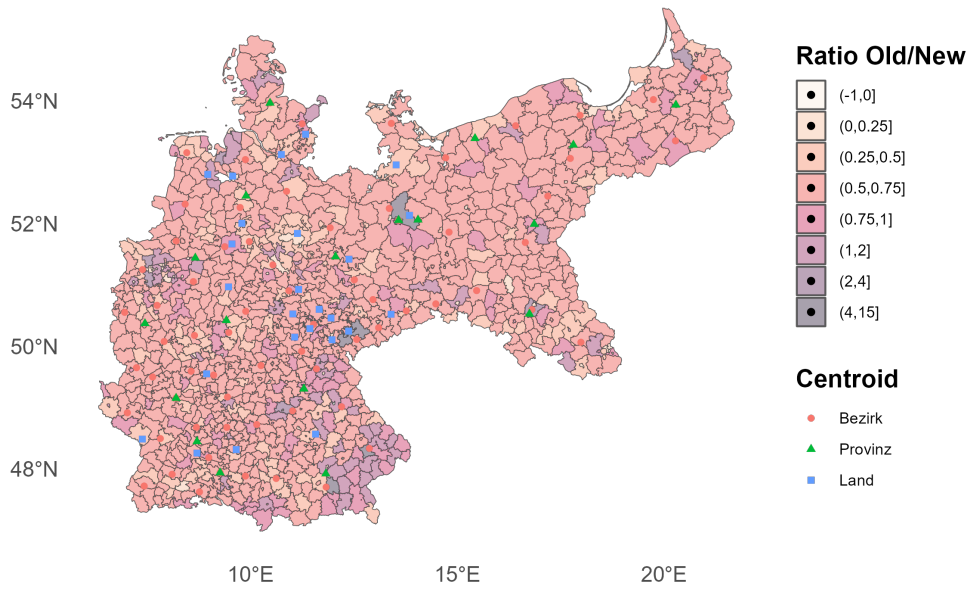
In the end, we were able to match around 7.5 million of the total 8.5 million entries to an entry at the fourth ("Kreis") or fifth ("Gemeinde") administrative level. Of the remaining entries, many do either not identify a location in Germany (e.g., "Philadelphia, Vereinigte Staaten") or only contain information on a higher administrative level (e.g., "Bayern") that does not allow matching it to the county level.

Our way of coding significantly improves the existing geocoding of casualty entries. The original data set by Verein für Computergenealogie (2019a) was automatically geocoded via the Historic Gazetteer as part of a Master Thesis by Sen (2016). While this automatic coding already delivered a very good spatial overview of casualty entries, it contained a number of structural errors that made the data difficult to use for our research purposes.

Our coding deviates in several ways from the original coding. In fact, the correlation between casualties per county according to our coding and the original coding is only 0.47. Figure A1 illustrates these coding disparities by plotting the casualties per county according to our coding over the casualties per county according to the original coding. Here, we first see three obvious cases where our coding leads to significantly more casualties than the old coding. Most obviously, Berlin sticks out with much more casualties according to our coding. We suspect that several Berlin-entries ended up in the neighboring district Niederbarnim (a significant outlier to the right), as the coordinates assigned to Berlin in the original coding were not accurately matching Berlin's administrative boundaries. Second, Frankfurt am Main and Mülheim an der Ruhr also deviate upwards. Here, we trace the difference to the original coding mistakenly sorting entries like "Frankfurt" to the (much less populated) "Frankfurt Oder" in Eastern Germany. Similarly, several cases for "Mülheim" were mistakenly allocated to Mühlheim am Rhein.

There are also several deviations towards the right hand side, i.e., where the

Figure A2: Geographic Differences in Geocoding



Note: This map illustrates the geographic variation in the differences in geocoded casualties. The color palette illustrates the ratio of a county's casualties in the original coding vs. our coding. The points depict the geographic centroids of higher administrative units, i.e., districts ("Bezirk"), provinces ("Provinz"), and states ("Land").

original coding suggests higher casualty counts as our coding. These deviations are better illustrated by the map in Figure A2. Here, dark-red to violet colors indicate counties where the ratio of casualties according to the original coding vs. our coding is between 2 and 15. In addition, we plot the centroids of higher administrative units, i.e., districts, provinces, and states. This shows that extraordinarily higher codings in the original geocoding tend to overlap with administrative centroids. For example, Niederbarnim, the large outlier in the scatterplot in the North-East close to Berlin, contains the centroids of both the province Brandenburg and the state of Prussia. The many entries which only held the information "Preußen" or "Brandenburg" were hence sorted into Niederbarnim. Similar cases are, e.g., Breslau, Düsseldorf, Ebersberg, and Landau an der Isar.

B Birthplaces as measure for places of living

Another concern that inherits from the nature of our data is internal migration. Unfortunately, the source of our data only identifies soldiers' birthplaces, not their place of residence before the war. While we lack the data for a systematic check of migration patterns, we can leverage the information on army districts to evaluate how likely internal migration is to introduce a severe bias into our regressions. Soldiers were assigned to military units according to their place of residence.[22] Therefore, the majority of soldiers' birthplaces should fall into their regiments' borders if the county of birth is identical to the county of residence, i.e., if pre-war migration among soldiers within German borders was low. In Figure B4 in the Appendix, we plot the casualty rates associated with the 49th Infantry Regiment based in the city of Gnesen in the province of Posen in the most South-Eastern part of the district as an example. The map shows that the vast majority of casualties lists birthplaces that lie within the borders of the respective army district, indicated by thick black lines. Indeed, this is the case for all list entries for which we have information on the associated army district.

Overall, the figure suggests that assigning casualties based on birthplaces gives a mostly accurate picture of where soldiers lived before the war. Even though internal migration, especially from rural towards urban areas, existed in Germany in the pre-war period (Grant, 2005) and we must acknowledge that soldiers' birthplaces make an imperfect proxy for their place of residence, we are confident that our somewhat noisy casualty measure does not introduce a structural bias into our regressions for two reasons. First, internal migration consisted largely of urban-rural movements. Our Difference-in-Differences identification strategy accounts for urban-rural differences, e.g., in migration rates/directions, via county fixed effects.

Second, our analysis aims to shed light on the local labor market effects of WWI casualties. A priori, it is unclear whether the place of residence or indeed the birthplace of a soldier constitutes the optimal metric here. Contemporary sources suggest that young men often only migrated temporarily. Especially soldiers returning from war often searched employment close to their parental homes. Hence, any impact from wounded or missing men is at least as likely to occur at soldiers' birthplaces as at their pre-war place of residence. We leverage an additional set of lists of missing soldiers that was collected after the war and digitized by the Verein für Computergenealogie (2019b). These lists contain soldiers' birthplaces along with the place of residence of their closest relatives, usually their parents or spouse. We match birthplaces and the residence of the closest relative and find that for about 60% of entries, the address of the closest relative was identical to the place of birth of the missing soldier, accounting only for slight differences in spelling and typos. Note that these 60% only include cases of exact address matches and still exclude cases where soldiers moved within the same village or town. Overall, we are therefore confident that soldiers' birthplaces serve as a valid proxy for the local impact of WWI casualties.

[22] As the number of casualties increased over time, this rule was relaxed significantly (Stachelbeck, 2010).

C Data and Descriptives

Table B1: Descriptive statistics

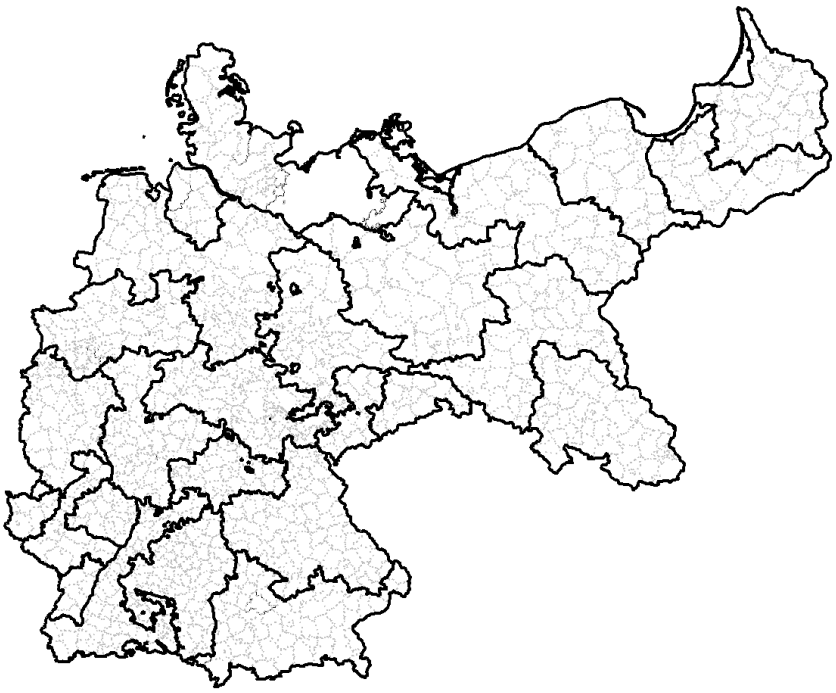
	Count	Mean	SD	Min	Max
Casualty rate	1,432	0.25	0.06	0.05	0.46
Death rate	1,432	0.10	0.03	0.02	0.26
Employment (log)	1,432	9.92	0.66	6.83	12.38
Primary employment (log)	1,432	9.31	0.99	5.64	10.97
Secondary employment (log)	1,432	8.27	1.14	0.00	11.69
Tertiary employment (log)	1,432	7.34	1.02	0.00	11.14
Employment: white collar (log)	1,432	7.77	0.99	0.00	11.26
Employment: blue collar (log)	1,432	8.06	1.19	0.00	11.59
White collar share	1,432	0.14	0.09	0.00	0.50
White collar share: secondary	1,431	0.32	0.10	0.05	0.72
White collar share: tertiary	1,431	0.72	0.11	0.19	0.96
Female share	1,432	0.04	0.06	-0.17	0.63
Population (log)	1,432	10.62	0.63	8.43	13.10
Wages, male over 21	1,431	13.01	10.89	1.60	50.00
Wages, male under 21	1,431	10.10	8.37	1.20	40.00
Wages, female over 21	1,431	9.17	7.78	1.00	40.00
Wages, female under 21	1,431	7.18	5.99	0.70	30.00
Primary employment: total (log)	1,432	9.29	1.04	3.76	10.97
Primary employment: subsistance farms (log)	1,432	7.92	1.08	3.53	10.33
Primary employment: small farms (log)	1,432	7.51	1.25	1.39	9.72
Primary employment: medium-sized farms (log)	1,432	8.02	1.26	1.61	9.75
Primary employment: large farms (log)	1,432	6.75	1.74	0.00	9.34
Primary employment: very large farms (log)	1,432	4.54	2.92	0.00	9.33
Number of all farms (log)	1432	8.29	0.95	4.11	10.15
Number of subsistance farms (log)	1,432	7.59	1.05	3.30	9.96
Number of small farms (log)	1,432	6.46	1.21	1.39	8.47
Number of medium-sized farms (log)	1,432	6.54	1.27	1.10	8.25
Number of large farms (log)	1,432	4.62	1.67	0.00	7.39
Number of very large farms (log)	1,432	1.85	1.54	0.00	5.34
Size of all farms (log)	1,432	9.80	1.27	3.71	11.78
Size of subsistance farms (log)	1,432	6.97	1.08	2.75	9.07
Size of small farms (log)	1,432	7.65	1.23	2.36	9.63
Size of medium-sized farms (log)	1,432	8.78	1.32	2.64	10.63
Size of large farms (log)	1,432	8.03	1.93	0.00	11.03
Size of very large farms (log)	1,432	5.78	3.45	0.00	11.09
Prussian county	1,432	0.55	0.50	0.00	1.00

Table B2: Sector structure of the business census

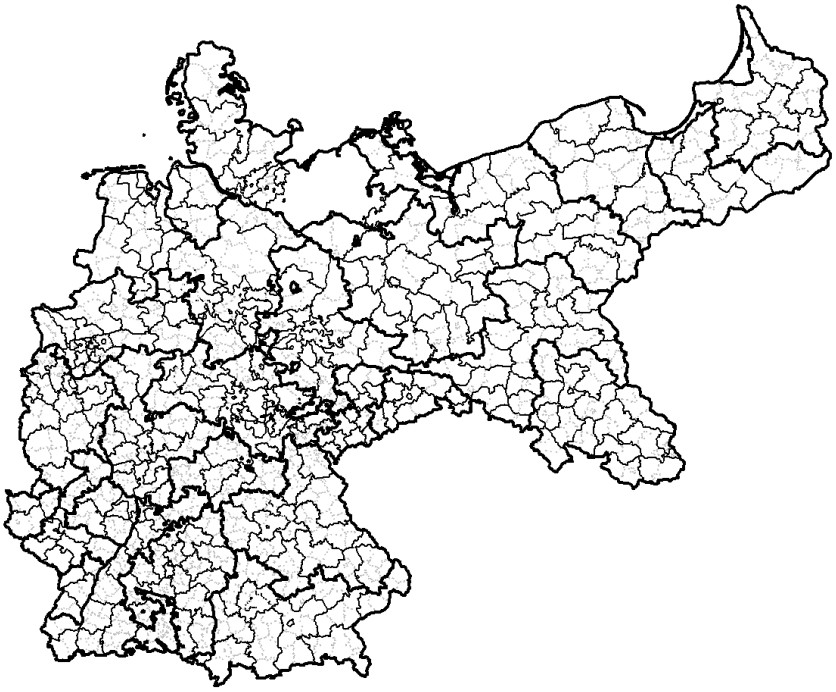
Sector	Subsector	Sub-subsector	Share	War related industry
Primary	Agriculture	Non-agricultural horticulture, animal husbandry, and deep-sea fishing	0.0154	.
Secondary	Mining	Mining, saltworks and peat digging	0.0755	x
Secondary	Mining	Industry of stones and earths	0.0763	
Secondary	Manufacturing	Manufacture of iron, sheet metal and metal ware	0.0634	x
Secondary	Manufacturing	Machinery, apparatus and vehicle construction	0.0604	x
Secondary	Chemical and Textile Industry	Chemical industry	0.0340	x
Secondary	Chemical and Textile Industry	Textile industry	0.1079	.
Secondary	Consumption goods	Paper industry and multiplying trades	0.0386	.
Secondary	Chemical and Textile Industry	Leather and linoleum industry	0.0205	.
Secondary	Constuction	Wood and cuttings industry	0.0481	.
Secondary	Consumption goods	Food and beverage industry	0.0524	.
Secondary	Consumption goods	Clothing industry	0.0467	.
Secondary	Constuction	Construction (incl. ancillary construction)	0.0446	.
Tertiary	Trade and insurance	Commerce	0.0493	.
Tertiary	Trade and insurance	Insurance	0.0059	.
Tertiary	Transit, gastronomy, other	Transport	0.0487	.
Tertiary	Transit, gastronomy, other	Catering and tavern industry	0.0420	.
Tertiary	Transit, gastronomy, other	Other	0.0176	.

Figure B1: German Empire, army districts, and military districts

a) Army districts

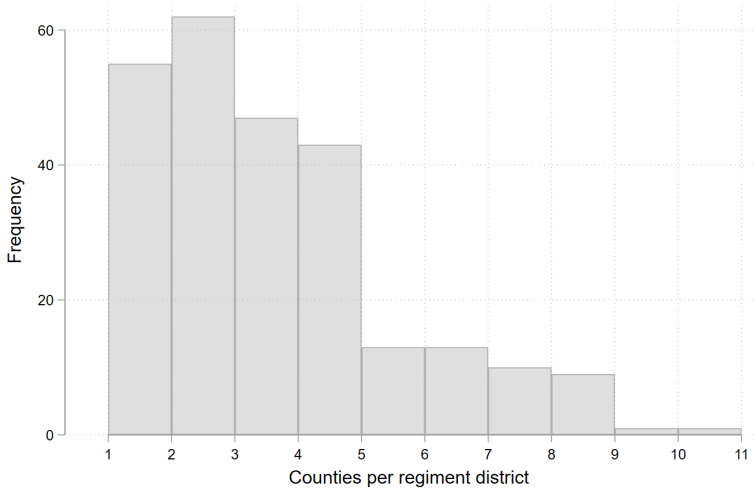


b) Military districts



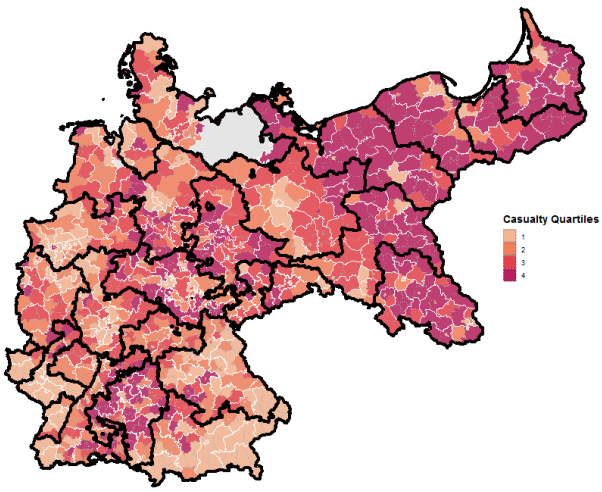
Notes: Figure B1 a) shows the 1914 German army districts ('Armeebezirke'). Thick black lines depict army district borders. Gray borders depict counties. Figure B1 b) shows the 1914 German regiment districs ('Landwehrbezirke'). Thick black lines depict army district borders. Thin black lines depict military district borders. Gray borders depict counties.

Figure B2: Number of counties per military district



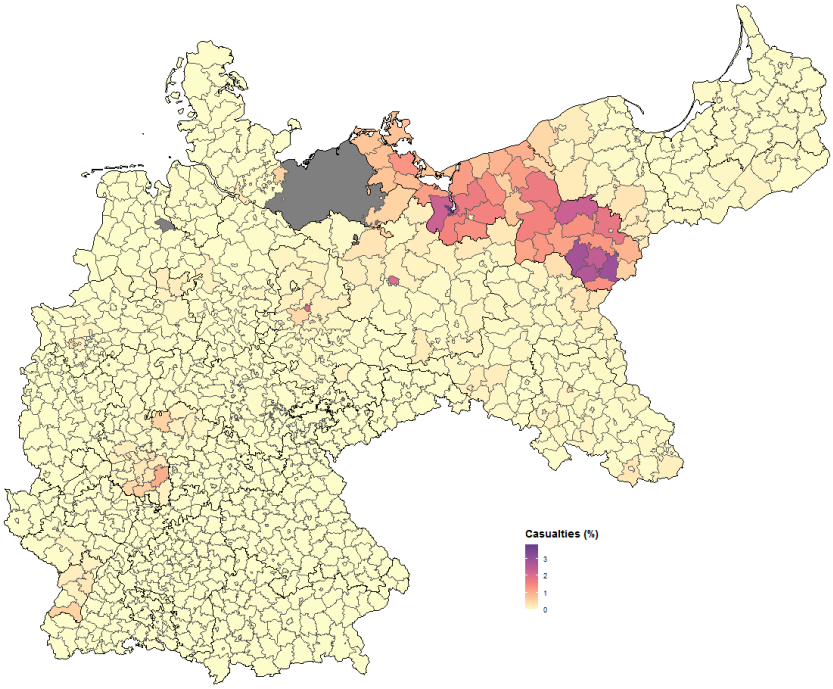
Note: This graph reports distribution of the number of counties within a military district.

Figure B3: Casualty quartiles



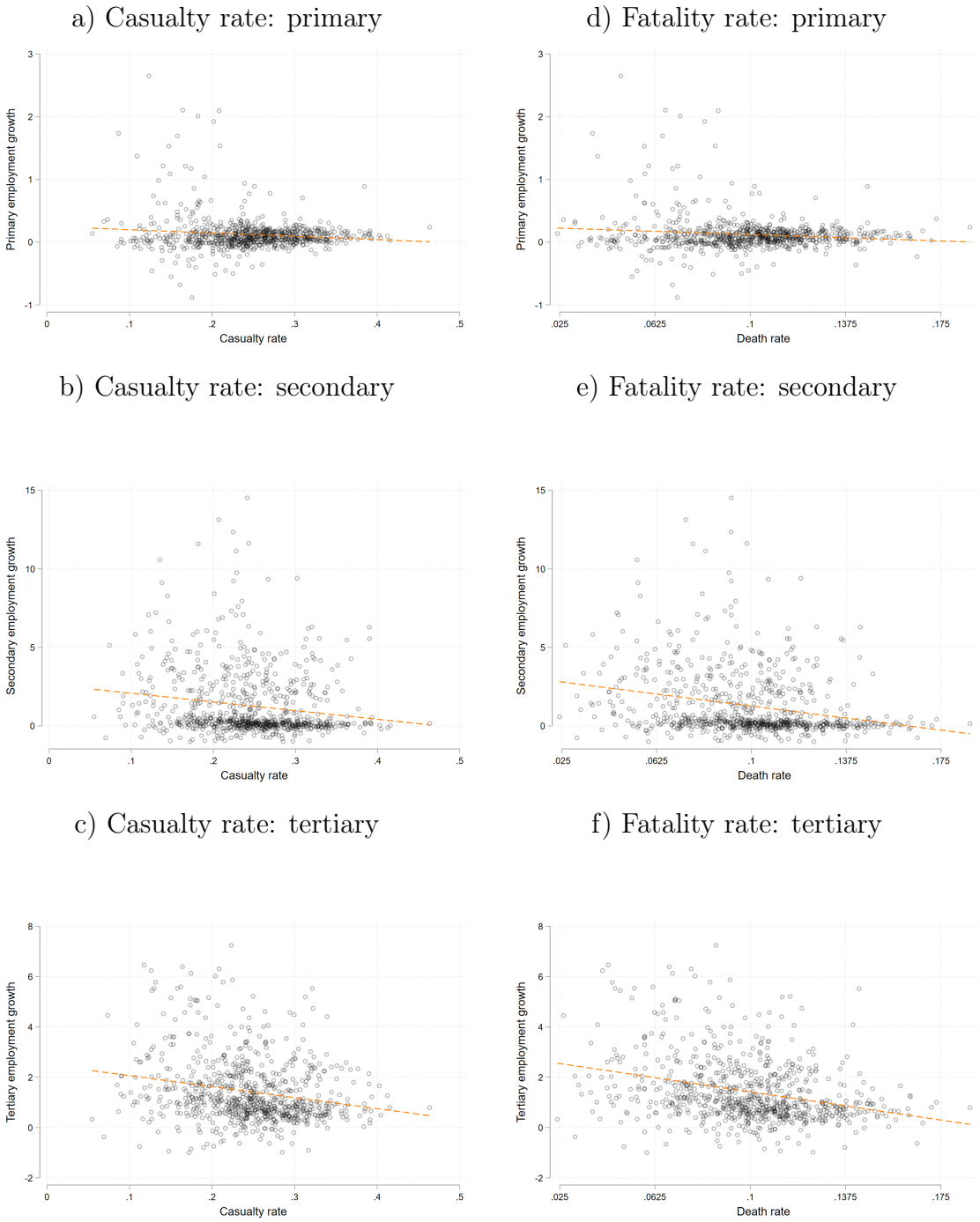
Notes: This figure illustrates the geographic distribution of the casualty quartiles for our identification of local marginal effects. Thick black lines depict army district borders. White borders depict military district borders. Thin dotted gray lines depict county borders. Grey areas denote missing data.

Figure B4: Birthplaces of casualties from the 49th Infantry Regiment



Notes: This map shows, for the 49th Infantry Regiment, which share of the regiment's total casualties fell into each German county. Thick black lines depict army district borders. Thin gray lines depict county borders. Grey areas denote missing data.

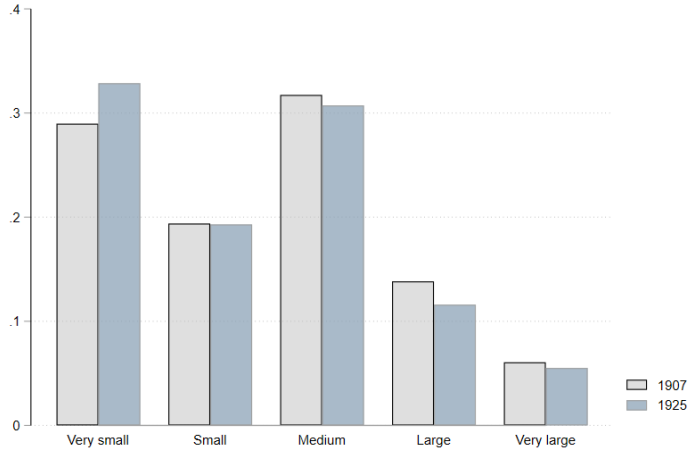
Figure B5: War exposure and sectoral employment growth



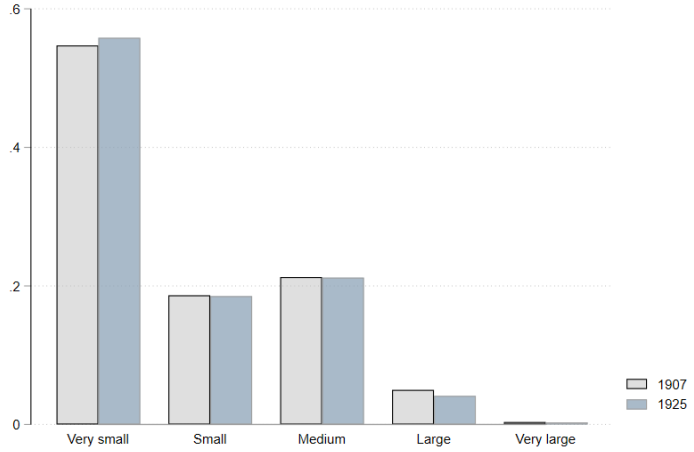
Notes: The graphs report correlations between war exposure and sectoral employment growth rate between 1910 and 1925. Figures a)–c) report correlations for the casualty rate, Figures d)–f) report correlations for the fatality rate. All counties in the final sample are included once.

Figure B6: Developments in the agricultural sector

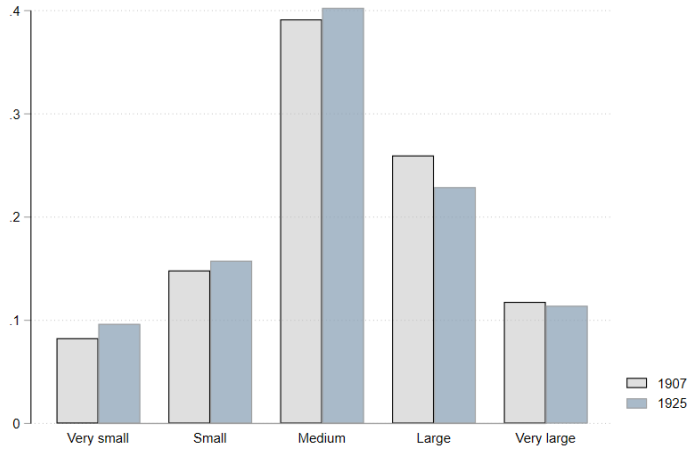
(a) Number of employees



(b) Number of farms



(c) Farm area (in hectares)



Notes: The graphs shows the share of agricultural employees (a), number of farms (b), and total cultivated area (c) by farm size in 1907 and 1925. "Very small" farms are farms with a total cultivated area of less than 2 ha, "small" farms between 2–5 ha, "medium" farms between 5–20 ha, "large" farms between 20–100 ha, and "very large" farms above 100 ha.

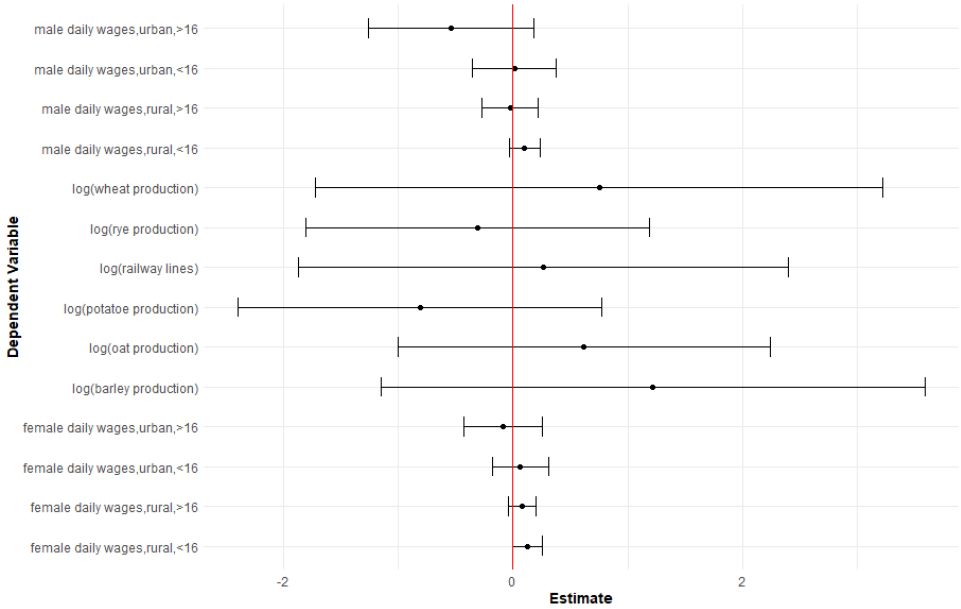
D Additional Tables and Figures

Table C1: Employment effects, pre-trends

	1867–1882 (1)	1882–1895 (2)	1895–1910 (3)	1910–1925 (4)
Panel A: primary sector				
Dead × Post	0.00048	0.00546	0.00470	-0.01773**
	(0.00720)	(0.00540)	(0.00505)	(0.00896)
Observations	524	524	524	518
Panel B: secondary sector				
Dead × Post	-0.00203	-0.00206	0.00082	-0.01540
	(0.00828)	(0.00630)	(0.00732)	(0.02121)
Observations	524	524	524	518
Panel C: tertiary sector				
Dead × Post	0.00512	-0.00108	-0.00031	-0.03401**
	(0.00910)	(0.00861)	(0.00506)	(0.01567)
Observations	524	524	524	518
County FE	✓	✓	✓	✓
Post x Regiment district	✓	✓	✓	✓
Post x Mobilization	✓	✓	✓	✓
Post x Quartiles	✓	✓	✓	✓

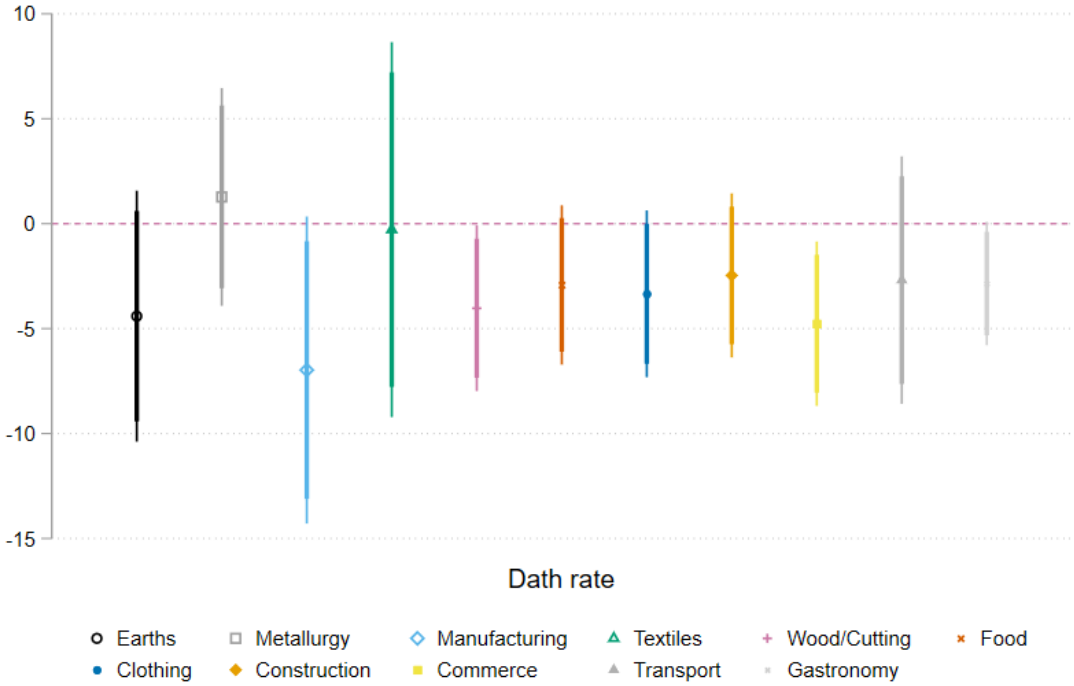
Notes: Results of Difference-in-Differences regressions following Equation 1. Outcome variables are the shares of a county's employment in agriculture (Panel A), the manufacturing industry (Panel B), and trade-related occupations (Panel C) over total employment. The main explanatory variable is the share of WWI casualties over the county's pre-war male population. All specifications control for logged population. Standard errors clustered by county: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure C1: Additional pre-trends



Note: This graph reports the coefficients and 95% confidence intervals from TWFE regressions. Dependent variables are the length of railway lines in 1856 and 1905, wages in 1892 and 1901, and agricultural production in 1886 and 1896 for Prussia.

Figure C2: Employment effects, sub-subsector results



Notes: Figure B1 a) shows the 1914 German army districts ('Armeebezirke'). Thick black lines depict army district borders. Gray borders depict counties. military district borders. Figure B1 b) shows the 1914 German regiment districs ('Landwehrbezirke'). Thick black lines depict army district borders. Thin black lines depict military district borders. Gray borders depict counties.

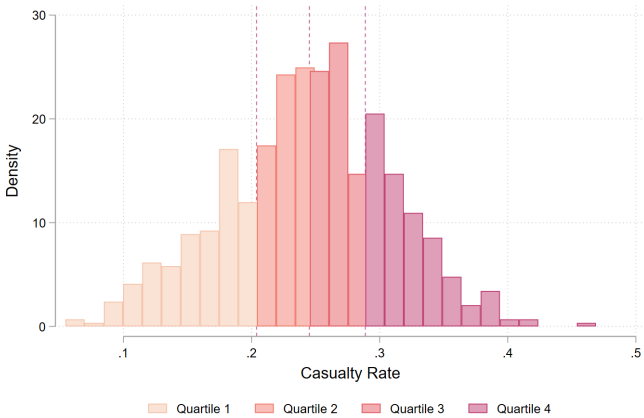
Table C2: WWI casualties and sectoral employment, subsector results

	Mining (1)	Manufacturing (2)	Chemicals (3)	Building (4)	Consumption (5)	Trade (6)	Transport (7)
Dead × post	-2.96453	-0.89470	4.65178	-2.29989	2.34267	-3.54951*	-4.77636**
	(1.97950)	(3.09057)	(3.62201)	(2.86052)	(4.63851)	(1.99006)	(1.91537)
Observations	1432	1432	1432	1432	1432	1432	1432
R^2	0.927	0.950	0.956	0.987	0.940	0.993	0.936
Mean DV	7.142	5.806	6.835	5.406	5.234	5.693	6.790
County FE	✓	✓	✓	✓	✓	✓	✓
Post x military district	✓	✓	✓	✓	✓	✓	✓
Post x female share	✓	✓	✓	✓	✓	✓	✓
Post x quartiles	✓	✓	✓	✓	✓	✓	✓

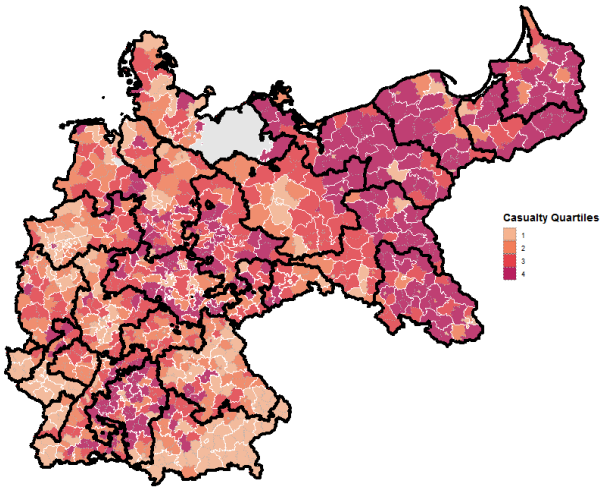
Notes: Results from continuous difference-in-differences estimations following equation 1. Outcome variables measure logged sectoral employment as indicated in column titles. "Casualties × post" indicates the share of casualties in the male 1910 population interacted with a dummy indicating the post-war period. "Dead × post" indicates the share of fatalities in the 1910 male population interacted with a dummy indicating the post-war period. All specifications control for logged population. Standard errors clustered by county in parentheses, Conley standard errors in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure C3: Casualty quartiles

a) Distribution of quartiles

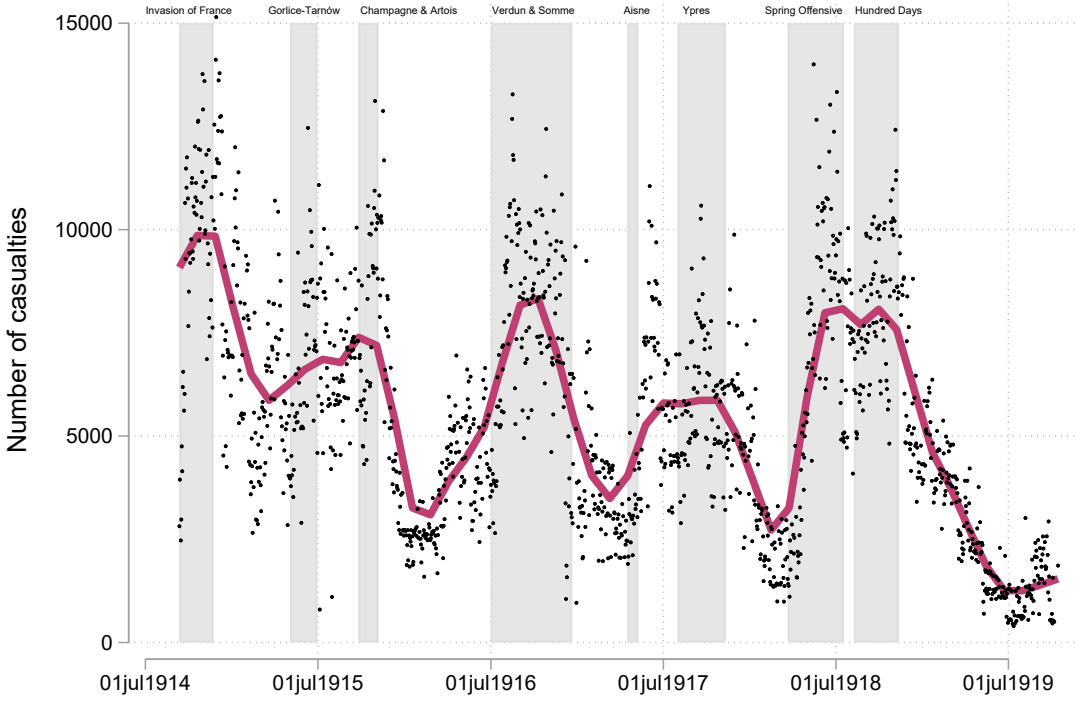


b) Spatial distribution of quartiles



Notes: These figures illustrate the empirical distribution of our main explanatory variable, the rate of WWI casualties in the pre-war 1910 male population, along with the allocation into quartiles for our identification of local marginal effects. Figure a) depicts the histogram of the casualty rate. Vertical lines and shades distinguish casualty rate quartiles. Figure b) depicts the allocation of counties into casualty rate quartiles on a map. Thick black lines depict army district borders. White borders depict military district borders. Thin dotted gray lines depict county borders. Grey areas denote missing data.

Figure C4: German WWI casualties and large battles



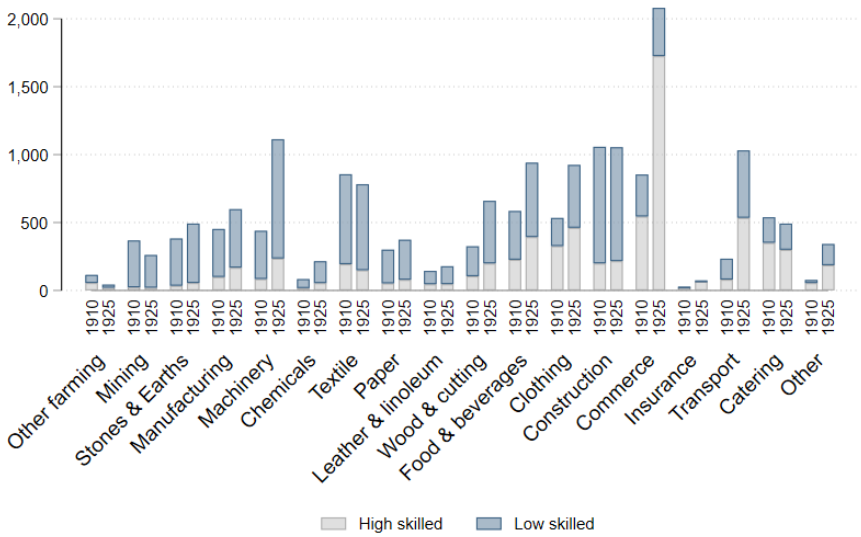
Note: This figure depicts the number of dead and wounded German WWI casualties in the German loss lists by day during WWI. Respective battles are shaded grey.

Table C3: WWI casualties, sectoral employment, and 1910 employment shares

	Employment: Primary (1)	Employment: Secondary (2)	Employment: Tertiary (3)	Employment: Primary (4)	Employment: Secondary (5)	Employment: Tertiary (6)	Employment: Primary (7)	Employment: Secondary (8)	Employment: Tertiary (9)
	Primary share (1910)			Secondary share (1910)			Tertiary share (1910)		
Dead × post	-1.69427**	-1.46889	-4.15021**	-1.50894*	11.15833***	-11.44524***	-2.51794***	-7.23822***	-0.71540
	(0.72263)	(2.04869)	(1.75474)	(0.83859)	(2.77887)	(2.19330)	(0.83198)	(2.51132)	(2.13591)
Interaction	-16.50275***	13.05397	-0.44304	-0.78018	-18.57006***	10.73506***	1.57216*	19.07892***	-11.33956***
	(3.92908)	(11.29440)	(9.20967)	(0.81043)	(3.47653)	(2.69992)	(0.86643)	(3.64010)	(2.78993)
Observations	1426	1426	1426	1432	1432	1432	1432	1432	1432
County FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Post x military district	✓	✓	✓	✓	✓	✓	✓	✓	✓
Post x female share	✓	✓	✓	✓	✓	✓	✓	✓	✓
Post x quartiles	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Results from continuous difference-in-differences estimations following equation 1. Outcome variables measure logged sectoral employment as indicated in column titles. "Casualties × post" indicates the share of casualties in the male 1910 population interacted with a dummy indicating the post-war period. "Dead × post" indicates the share of fatalities in the 1910 male population interacted with a dummy indicating the post-war period. All specifications control for logged population. Standard errors clustered by county in parentheses, Conley standard errors in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure C5: Employment and skill level by sub-sector, 1910 and 1925



Notes: This figure depicts the share of high-skilled workers ("Angestellte") and low-skilled workers ("Arbeiter") in total employment by subsector according to the 1910 and 1925 business census.

Table C4: WWI casualties and employment, subsample analysis

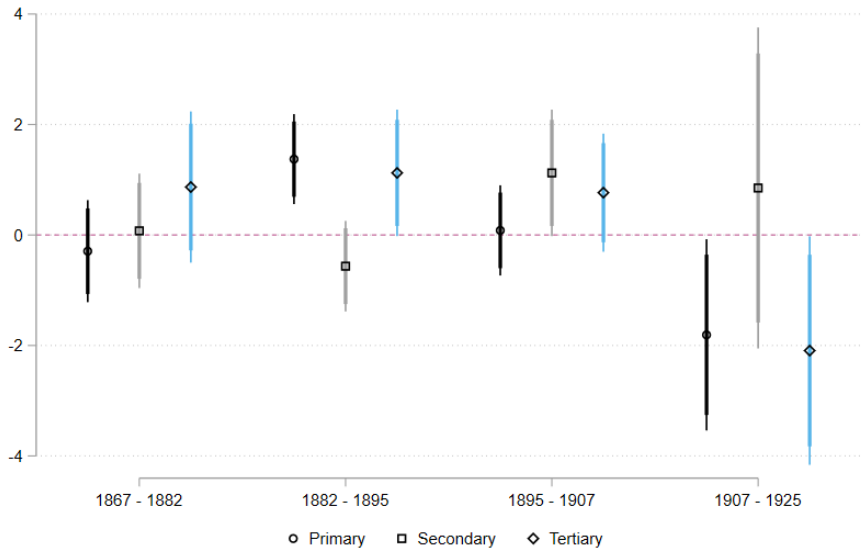
Dropped subsample:	Baseline	Ruhr area	Russian advance	Non-Galloway	Top 10%	Bottom 10%
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Employment						
Dead × post	-4.65428	-5.17785	-4.81723	-1.55133	-3.19806	-8.65803
	(1.27049)***	(1.27159)***	(1.31245)***	(0.61965)**	(1.04608)***	(2.34506)***
	[1.51514]***	[1.69846]***	[1.24322]***	[0.25770]***	[1.12444]***	[2.21589]***
Panel B: Employment: Primary						
Dead × post	-2.03624	-1.84684	-2.14724	-1.83605	-1.63581	-3.09440
	(0.74355)***	(0.74108)**	(0.77378)***	(0.93189)**	(0.90500)*	(1.10358)***
	[0.58249]***	[0.58030]***	[0.64496]***	[0.32530]***	[0.74179]**	[0.36899]***
Panel C: Employment: Tertiary						
Dead × post	-4.18979	-4.57117	-4.33323	-3.80464	-5.58000	-4.38710
	(1.74993)**	(1.73070)***	(1.79314)**	(1.52294)**	(1.85512)***	(2.94766)
	[0.94258]***	[1.08596]***	[0.97862]***	[0.90408]***	[1.52174]***	[1.73378]**
County FE	✓	✓	✓	✓	✓	✓
Post x regiment district	✓	✓	✓	.	✓	✓
Post x mobilization	✓	✓	✓	✓	✓	✓
Post x quartiles	✓	✓	✓	✓	✓	✓
Observations	1,432	1,268	1,374	518	1,284	1,250

Notes: Results from continuous difference-in-differences estimations following equation 1, using subsamples. Column 1 shows the baseline results. Column 2 drops counties affected by the 1923 French and Belgian occupation of the Ruhr area. Column 3 drops the territory in East Prussia invaded by Russia in 1914. Column 4 drops all non-Galloway counties. Column 5 drops counties in the highest decile of war exposure. Column 6 drops counties in the lowest decile of war exposure. Standard errors clustered by county in parentheses, Conley standard errors in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We find that daily laborers' wages grew slower in counties with higher casualties, whereas we do not find significant evidence for wage reactions to WWI fatalities. For adult male worker over the age of 21, a one standard deviation increase in casualties led to about one Reichsmark lower wage growth. Compared to an average wage increase in the low-skilled sector of around 20 Reichsmark, this is a quite sizeable effect, especially considering that it is likely the outcome of two opposing effects. On the one hand, one would a priori expect wages to rise in counties with higher casualties or fatalities, as labor becomes scarce. Yet on the other hand, one might expect a selection effect taking place if businesses recruit the most able daily laborers to fill vacancies in the formal Blue Collar sector. This selection effect would lead low-skilled wages to deteriorate as the average skill level among the remaining workers decreases. Our results are in line with both of these effects taking place, while both effects cancel each other out when looking only at fatalities, but the selection effect dominating when also accounting for wounded soldiers. Likely, wounded veterans that return and join the daily laborer market lead to a further decrease in the average skill level, allowing us to identify statistically significant wage effects.

Overall, we interpret the results from Table C7 as evidence that businesses also recruited Blue Collar workers from the low-skilled sector. If such recruitment didn't take place, theory dictates that we should observe a significant increase in wages amid higher labor scarcity in harder hit counties. Our results let us rule out higher

Figure C6: WWI casualties and structural transformation, pre-trends without casualty quartile fixed effects



Notes: This figure depicts the coefficients as well as their 90%- and 95% confidence intervals from the test for pre-trends by estimating the continuous difference-in-differences regressions as outlined in equation 1, but without casualty quartile fixed effects and for the periods 1867–1882, 1882–1895, and 1895–1907 for the harmonized sample of Prussian counties in the Galloway data. "Primary" indicates logged employment in the agriculture sector, colored in black. "Secondary" indicates logged employment in the manufacturing sector, colored in gray. "Tertiary" indicates logged employment in the trade and service sector, colored in blue. We report results by pairs of years for the three sectors as indicated on the X-axis. All coefficients derive from separate regressions. All specifications include army district fixed effects, the female share, and logged population. Data before 1910 derive from Galloway (2007).

Table C5: WWI casualties and employment, alternative definition of treatment doses

	Employment (1)	Employment: Primary (2)	Employment: Secondary (3)	Employment: Tertiary (4)
Panel A: Medians				
Dead × post	-5.19283	-2.59129	-0.64991	-3.93845
	(0.92725)***	(0.66434)***	(1.94713)	(1.68163)**
Panel B: Terciles				
Dead × post	-4.28395	-2.25363	-0.77191	-1.48227
	(1.09024)***	(0.70298)***	(2.58860)	(2.30673)
Panel C: Quintiles				
Dead × post	-3.71983	-1.78176	-0.33601	-1.87538
	(1.39304)***	(0.71836)**	(2.37227)	(2.08631)
Panel C: Sixtiles				
Dead × post	-3.16548	-2.65410	-2.92409	-2.68993
	(1.39168)**	(0.95067)***	(2.28291)	(2.20605)
County FE	✓	✓	✓	✓
Post x regiment district FE	✓	✓	✓	✓
Post x mobilization	✓	✓	✓	✓
Post x treatment dose FE	✓	✓	✓	✓
Observations	1432	1432	1432	1432

Notes: Results from continuous difference-in-differences estimations following equation 1. Each panel uses a different definitoin of hte casualy quartiles, as indicated. Outcome variables measure logged total employment (column 1), and logged employment in the primary, secondary, and tertiary sector (columns 2–4). "Dead × post" indicates the share of fatalities in the 1910 male population interacted with a dummy indicating the post-war period. All specifications control for logged population. Standard errors clustered by county in parentheses, Conley standard errors in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

wage increases in harder hit counties with almost certainty. Hence, we can summarize the war-induced transformation of the German labor markets as follows: Manufacturing firms recruited labor from the agricultural and informal sector to rapidly replace the Blue Collar workers lost in WWI. Firms in the tertiary sector that relied more on high-skilled White Collar workers lacked this labor reserve and therefore had to leave many positions vacant.

Table C6: WWI casualties and employment, alternative mobilization proxies

	Employment (1)	Employment: Primary (2)	Employment: Secondary (3)	Employment: Tertiary (4)
Panel A: Female growth rate (baseline)				
Dead × post	-4.65428	-2.03624	-1.39253	-4.18979
	(1.27049)***	(0.74355)***	(2.05606)	(1.74993)**
Panel B: Difference in female share, 1916 - 1910				
Dead × post	-4.04848	-1.95318	-1.38690	-4.06034
	(1.19742)***	(0.73955)***	(2.10325)	(1.78387)**
Panel C: Simple female share, 1916				
Dead × post	-5.18382	-2.17723	-1.64061	-4.57009
	(1.33523)***	(0.76458)***	(2.02153)	(1.72602)***
County FE	✓	✓	✓	✓
Post x regiment district FE	✓	✓	✓	✓
Post x mobilization	✓	✓	✓	✓
Post x treatment dose FE	✓	✓	✓	✓
Observations	1432	1432	1432	1432

Notes: Results from continuous difference-in-differences estimations following equation 1. Each panel uses a different definitoin of hte casualy quartiles, as indicated. Outcome variables measure logged total employment (column 1), and logged employment in the primary, secondary, and tertiary sector (columns 2–4). "Dead × post" indicates the share of fatalities in the 1910 male population interacted with a dummy indicating the post-war period. All specifications control for logged population. Standard errors clustered by county in parentheses, Conley standard errors in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C7: Employment effects, wages

	Wage (m) (1)	Wage (f) (2)	Wage (m) under 21 (3)	Wage (f) under 21 (4)
Panel A: Casualties				
Casualties × post	-15.78624	-11.45959	-9.36177	-4.53131
	(7.50208)**	(5.91481)*	(5.63858)*	(4.47740)
	[6.22468]***	[5.09007]**	[5.35730]*	[4.33234]
Mean DV	13.007	10.099	9.162	7.174
Panel B: Dead				
Dead × post	-21.98314	-19.52155	-11.09736	-4.97264
	(15.29119)	(12.00273)	(10.83567)	(8.52181)
	[13.99224]*	[11.20067]*	[11.17595]	[8.15955]
Mean DV	13.007	10.099	9.162	7.174
County FE	✓	✓	✓	✓
Post x military district	✓	✓	✓	✓
Post x Female Share	✓	✓	✓	✓
Post x Quartiles	✓	✓	✓	✓
Observations	1,450	1,450	1,450	1,450

Notes: Results from Continuous Difference-in-Differences estimations following Equation 1. Outcome variables are the average male wage (Column 1) and average female wage (Column 2), as well as both respective wages for young workers under 21 in Columns (3) and (4), respectively. "Mean DV" provides the mean of each dependent variable. All specifications control for logged population. "Casualties × post" indicates the share of casualties of the male 1907 population interacted with a dummy indicating the post-war period. "Dead × post" indicates the share of dead soldiers of the 1907 male population interacted with a dummy indicating the post-war period. Standard errors clustered by county in parentheses, Conley standard errors using a 100km distance-cutoff in brackets: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.